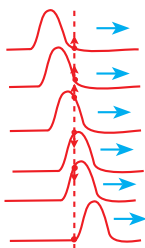


तो चलिए शुरू करते हैं wave motion बोले तो wave motion..... इस chapter को पढ़ने का अपना एक अलग ही तरीका है अगर उसके according चलोगे तो this chapter is high scoring chapter.... तरीका ये है किसी भी derivation पर जरूरत से ज्यादा ध्यान नहीं देना है और formula और result याद रखना है.... जितना content मैं यहाँ cover कर रहा हूँ तो Mains/Advance/NEET सबके सवाल बन जाएँगे।

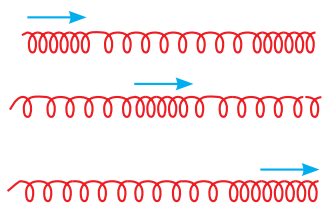


INTRODUCTION TO WAVES

A wave is a disturbance that propagates in space, transports energy and momentum from one point to another point without actual transport of matter particle.



(a)



(b)

Types of Waves

- (i) **Mechanical Waves:** Require material medium (elasticity and inertia) for their propagation. Example: Sound wave, Wave on string.
- (ii) **Electromagnetic or non-mechanical Waves:** Do not require any material medium for their propagation. Elasticity or inertia of medium do not affect the propagation of such waves. Example: Light and TV signals.

Classification of mechanical wave based on direction of motion of particle.

Transverse wave

Particle vibrates in perpendicular direction to the wave velocity.
ex. wave on string

Longitudinal wave

Particle vibrates in the direction of wave velocity (or in opposite direction).
ex. sound wave

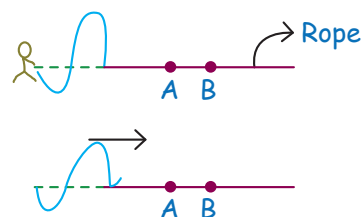
★ Essential property for mechanical wave propagation are inertia and elasticity.

Wave on String (Transverse Wave)

Particle vibrates in perpendicular direction to wave velocity.

Equation of Wave (Transverse)

Man gives disturbance in vertical direction to rope. And a wave has started to move as shown in figure.



यहाँ आदमी ने रस्सी पकड़ी है अगर आदमी के हाथ का motion बिल्कुल SHM of a particle जैसा है तो रस्सी में जो wave develop होगी वो SHM वाली transfer wave होगी जिसकी eqn. we can write as

$$y = A \sin(\omega t \pm kx + \phi)$$

ये eqn कैसे आई इसका derivation imp नहीं है, इस eqn में कौनसी physical quantity क्या है ये समझना बहुत imp है।

Equation of transfer wave is $y = A \sin(\omega t \pm kx + \phi)$ where

$y \Rightarrow$ Displacement of particle from mean position

$A \Rightarrow$ Amplitude

$\lambda \Rightarrow$ Wave length

$T \Rightarrow$ Time period of any particle of the string

$f \Rightarrow$ Frequency

$\omega \Rightarrow$ Angular frequency $\omega = 2\pi/T$

$k \Rightarrow$ Propagation const $k = \frac{2\pi}{\lambda}$

$\phi \Rightarrow$ Initial phase

Wave velocity (V_w) = $\frac{\omega}{k} = \frac{\text{Coeff. of } t}{\text{Coeff. of } x}$ (Magnitude)

★ अगर x और t के coeff का sign opposite है तो wave velocity $V_w \Rightarrow +x$ direction में

★ अगर x और t के coeff का sign same है $V_w = -x$ direction

Equation	Particle Velocity	Wave Speed
$y = A \sin(\omega t - kx + \phi)$	Along y	$+x$
$y = A \sin(\omega t + kx + \phi)$	y	$-x$
$y = A \sin(-\omega t + kx + \phi)$	y	$+x$
$y = A \sin(-\omega t - kx + \phi)$	y	$-x$

देख भाई wave की feel लेने के लिए मैं नीचे कुछ ques attach कर रहा हूँ जो आपके basic fundamental के लिए बहुत जरूरी है अब ये मत बोलना sir feel क्यों दिला रहे हो



Q. Equation of wave on string is represented by eqⁿ
 $y = 10 \sin(2\pi t - \pi x + \pi/3)$, Answer following parts.

Sol. (1) Amplitude (A) = 10 m

$\Rightarrow$ Angular frequency $\omega = 2\pi$ rad

$\Rightarrow$ Time period (T):

$$\therefore \omega = \frac{2\pi}{T} \Rightarrow T = 1 \text{ sec}$$

$\Rightarrow$ Frequency (f):

$$\therefore \omega = 2\pi f \Rightarrow f = 1$$

$\Rightarrow$ Wavelength (λ):

$$\therefore k = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{2\pi}{k}$$

$$\lambda = 2 \text{ m}$$

$\Rightarrow$ Initial phase $\phi = \pi/3$

$$(2) \text{ Wave speed} = \frac{\omega}{k} = \frac{2\pi}{\pi} = 2 \text{ m/s}$$

$$\Rightarrow \text{Wave velocity} = 2 (+x \text{ axis}) = 2\hat{i}$$

(3) Find equation of motion of particle at $x = 0$

$$y = 10 \sin(2\pi t + 0 + \pi/3)$$

$$y = 10 \sin(2\pi t + 60^\circ) \text{ SHM}$$

भाई कुछ नहीं करना बस wave की eqn में $x = 0$ put कर दो तो $x = 0$ वाले particle के motion की eqn आजाएगी.... अब ध्यान से देखो ये तो SHM की eqn आ गई समझे दया....



(4) Maximum speed of any particle

$$V_{\max} = A\omega = 10 \times 2\pi = 20\pi \text{ m/s}$$

(5) Find equation of motion of particle at $x = 1/6$

$$y = 10 \sin(2\pi t - \pi \cdot 1/6 + \pi/3)$$

$$= 10 \sin(2\pi t + \pi/6)$$

(6) Find location of particle at $x = 1/6$ at $t = 2$ sec

$$y = 10 \sin(2\pi \cdot 2 - \pi \cdot 1/6 + \pi/3) = 10 \sin(4\pi + \pi/6)$$

$$y = 10 \sin(\pi/6) = 5 \text{ m}$$

Q. In previous problem find speed of particle when it is at a distance 6 m from the mean position and its acceleration.

$$\text{Sol. } V = \omega \sqrt{A^2 - x^2}$$

$$= 2\pi \sqrt{(10)^2 - 6^2} = 16\pi \text{ m/s}$$

$$a = x\omega^2 = 6 \times (2\pi)^2 = 24\pi^2 \text{ m/s}^2$$

PROOF OF EQUATION OF WAVE

Equation of a Progressive Harmonic Wave

If the left end ($x = 0$) of the string is made to vibrate in simple harmonic motion, the equation of motion of this end can be expressed as

$$f(t) = A \sin(\omega t + \delta) \quad \dots (i)$$

The time period of oscillation is $T = \frac{2\pi}{\omega}$ and frequency

of oscillation $\nu = \frac{1}{T} = \frac{\omega}{2\pi}$. The wave produced by such

a vibrating source is called 'sine' wave or 'sinusoidal' wave given that the displacement of the particle at $x = 0$ is given by equation (i), the displacement of the particle at x at time t can be expressed as

$$y = f(t - x/v) = A \sin(\omega(t - x/v) + \delta)$$

$$= A \sin\left(\omega t - \frac{\omega}{v}x + \delta\right)$$

$$= A \sin(\omega t - kx + \delta)$$

where $k = \omega/v$ is called wave number. The velocity of the particle at x at time t is given by

$$\frac{\partial y}{\partial t} = A\omega \cos(\omega t - kx + \delta) \quad \dots (ii)$$

Please note that velocity of the particle is different from wave velocity v . The wave moves along x -axis at constant velocity v but particles move up and down

with velocity $\frac{\partial y}{\partial t}$ which depends on both position and time. The maximum velocity of the particle of the medium is ωA .

Equation of Wave (Transverse)

A travelling wave satisfies a differential equation, called the linear wave equation,

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} \quad \text{functions such as}$$

$$(i) A \sin(kx - \omega t), (ii) \frac{A}{B^2 + (x - vt)^2}, (iii) [A \sin$$

$$(kx - \omega t) + B \cos(kx + \omega t)], (iv) \sqrt{(ax + bt)}, (v) (ax -$$

$$bt)^2, (vi) Ae^{-B(x-vt)^2} \text{ or } (vii) A \cos^2(kx - \omega t) \text{ satisfy}$$

the wave equation, Practically wave equation is only valid when displacement is small. Hence, only (i), (ii), (iii), (vi) are valid wave functions but (iv) and (v) are not acceptable since they are unbounded.

Function should be

1. Defined
2. Continuous
3. Bound
4. x और t में आपस में relation में linear relation होना चाहिए।

Which of the following function can represent transverse wave equation

$$1. y = A \sin(\omega t - kx + \phi) \longrightarrow \text{Yes}$$

$$2. y = A \sin(\omega t^2 - kx + \phi) \longrightarrow \text{No}$$

$$3. y = A \sin(\omega t - kx^2 + \phi) \longrightarrow \text{No}$$

For a function to be wave function, the three quantities x , t and v must appear in the linear combinations $(x - vt)$ or $(x + vt)$. Thus, $(x - vt)^2$ is acceptable but $(x^2 - v^2t^2)$ is not (because it is not linear).

$$4. y = A \log(t^2 + x) \longrightarrow \text{No}$$

$$5. y = \frac{1}{(x + t)^2 - 4} \longrightarrow \text{No}$$

$$6. y = \frac{1}{(x + t)^2 + 4} \longrightarrow \text{Yes}$$

$$7. y = \frac{1}{x^2 + t^2 - 2xt + 10} \longrightarrow \text{Yes}$$

$$8. y = e^{-(ax + bt)^2} \longrightarrow \text{Yes}$$

In above question

$$\Rightarrow V_w = - \frac{\text{coeff of } t}{\text{coeff of } x} = \frac{-b}{a}$$

$$\longrightarrow \frac{b}{a} (-x \text{ में})$$

Q. Equation of transverse wave is given by

$$y = e^{-(ax^2 + bt^2 + 2\sqrt{ab}xt)}, \text{ find } V_w?$$

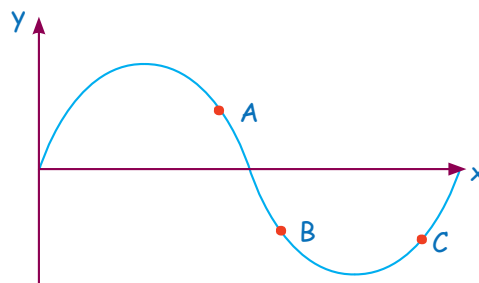
Sol. Above eqn can be written as $y = e^{-(\sqrt{a}x + \sqrt{b}t)^2}$

$$\text{Speed of wave } V_w = \frac{\text{coeff of } t}{\text{coeff of } x} = \frac{\omega}{k} (\text{magnitude})$$

$$V_w = \frac{\sqrt{b}}{\sqrt{a}} (-x \text{ axis})$$

Relation Between Wave Velocity (V_w) & Particle Velocity (V_p)

A wave is shown in figure



wave equation is

$$y = A \sin(\omega t - kx + \phi)$$

velocity of particle

$$V_p = \frac{dy}{dt} = A\omega \cos(\omega t - kx + \phi) \quad \dots(i)$$

$$\frac{dy}{dx} = -Ak \cos(\omega t - kx + \phi) \quad \dots(ii)$$

From eqn (i) & (ii)

$$\frac{V_p}{\left(\frac{dy}{dx}\right)} = \frac{A\omega \cos(\omega t - kx + \phi)}{-Ak \cos(\omega t - kx + \phi)}$$

$$\frac{V_p}{(dy/dx)} = \frac{\omega}{k} = -V_w$$

★ $V_p = -V_w \frac{dy}{dx}$ here, $\frac{dy}{dx} = \text{slope}$

★ $V_p = -V_w (\text{Slope})$

भाई यहाँ दो प्रकार की velocity है।

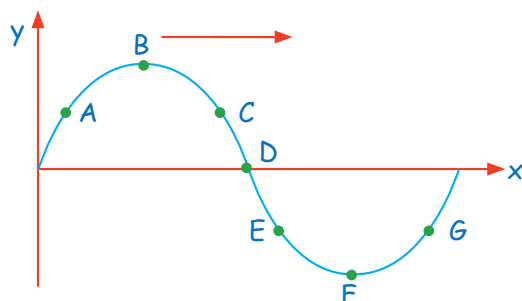
एक wave velocity ($V_w = \omega/k$) मतलब जिस velocity से wave दोड़ रही है।

और दूसरी particle velocity मतलब कौन-सा particle किस time पर किस velocity से move कर रहा है। ($V_p = \frac{dy}{dt}$)

यहाँ बिल्कुल confuse मत होना।



Q. If wave is moving along positive x direction and at any time t, relation between y vs x is given in following graph. Find direction of velocity of particle A, B, C, D, E, F, G.



Sol. किसी भी particle के लिए बस $V_p = -V_w (\text{slope})$ apply करना है। यहाँ V_w is positive for any particle.

For (A) $\Rightarrow V_p = -++$

$V_p < 0$ A के velocity नीचे है

For (B) $\Rightarrow \text{Slope} = 0$

$V_p = 0$ (extreme पर है)

For (C) $\Rightarrow V_p = -V_w (\text{slope})$

$V_p = -+- = +ve$

$V_p > 0$ उपर जा रहा है

For other

A, G $\Rightarrow \text{slope} > 0 \Rightarrow V_p < 0$

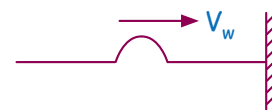
B, F $\Rightarrow \text{slope} = 0 \Rightarrow V_p = 0$

C, D, E $\Rightarrow \text{slope} < 0 \Rightarrow V_p > 0$

Wave Velocity in a String

Suppose a wave is moving in a string then speed of the wave can be written as

$$V_w = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\rho A}}$$



Here T = Tension in string

μ = Mass per unit length of string

ρ = Density of string

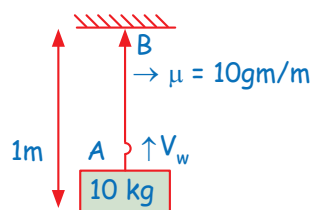
A = Area of cross section of string



कदू gang के वासियो इसका derivation बिल्कुल imp नहीं है so, derivation पे ध्यान मत देना formula पर ध्यान देना और ये Mains का one of the favourite article है जिसपर formula oriented बहुत सवाल पूछे गए हैं।

Q. A block of mass 10 kg is hanging by a string of mass per unit length 10 gm/m and of length 1 m. A small pulse is given from the lower end. Find velocity of wave V_w .

Sol.



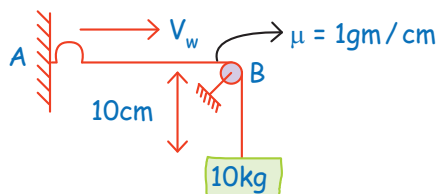
$T = Mg$ (every where)

$T = 100$ approx

$$V_w = \sqrt{\frac{T}{\mu}}$$

$$V_w = \sqrt{\frac{100}{10 \times 10^{-3}}} = 100 \text{ m/s}$$

- Q. In the following diagram a wave is moving from point A to B. Find V_w & time taken to travel from A to B. ($AB = 2m$).



Sol. $T = 100$ (approx)

$$\mu = \frac{1 \text{ gm}}{\text{cm}} = \frac{10^{-3} \text{ kg}}{10^{-2} \text{ m}} = 0.1 \text{ kg/m}$$

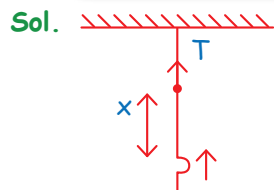
$$\text{Velocity of wave} = V_w = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{100}{0.1}} = 10\sqrt{10}$$

time to travel A to B

$$t_{A \rightarrow B} = \frac{AB}{V_w} = \frac{2}{10\sqrt{10}}$$

- Q. String of length L and mass m is hanging from a ceiling. If a wave start moving from lower end. Find V_w at a distance x from the bottom & time taken by wave to move from lower end to upper end.

अबे ये तो माल वाली रस्सी है यहाँ ये याद रखना
pulse का acc $g/2$ upward होगा।



$$\text{Tension at a distance } x \text{ from bottom} = T = \frac{m}{L} xg$$

$$\mu = \frac{m}{L}$$

$$V_w = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\frac{m}{L} xg}{m/L}} = \sqrt{xg}$$

$$V_w^2 = xg$$

$$2V_w \frac{dV_w}{dx} = g$$

$$\text{Acceleration } a = V_w \frac{dV_w}{dx} = \frac{g}{2} = \text{constant}$$

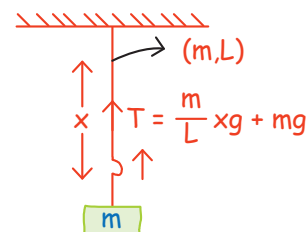
Time to reach up $t(A \rightarrow B) = ?$

$$L = 0 + \frac{1}{2} \left(\frac{g}{2} \right) t^2$$

$$t = 2\sqrt{\frac{L}{g}}$$

As pulse move up $x \uparrow$,
 $T \uparrow, V \uparrow, \therefore V = f\lambda$
 $V \uparrow \Rightarrow f$ same and $\lambda \uparrow$

- Q. Repeat the above problem if a block of same mass m is attach at the lowest point.



$$\text{Sol. } V_w = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\frac{m}{L} xg + mg}{m/L}}$$

$$V_w = \sqrt{xg + Lg}$$

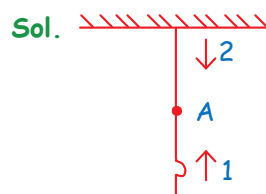
$$V_w^2 = xg + Lg$$

diff wrt x

$$2V_w \frac{dV_w}{dx} = g + 0$$

$$a = \frac{g}{2} \text{ (same)}$$

- Q. Suppose two pulse is given in a string (m, l) one from upper side and one from lower side as shown in fig. Compare the speed of wave 1 & 2 at A.



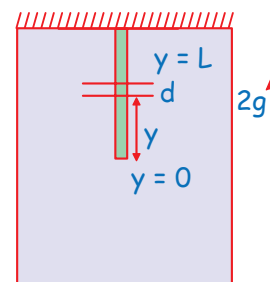
Sol.

$$(\text{Speed})_1 = (\text{Speed})_2$$

$$(\text{Velocity})_1 \neq (\text{Velocity})_2$$

- Q. One end of a string of length L is tied to the ceiling of a lift accelerating upwards with an acceleration $2g$. The other end of the string is free. The linear mass density of the string varies linearly from 0 to λ from bottom to top.

Sol.



Linear mass density of string, (μ)

$$\mu = \frac{\lambda}{L} y$$

$$\mu = \frac{dm}{dy} \Rightarrow dm = \mu dy$$

Mass of string of length y units is

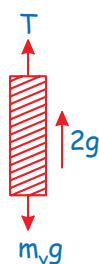
$$\int_0^{m_y} dm = \int_0^y \mu dy = \int_0^y \frac{\lambda}{L} y dy = \frac{\lambda}{L} \frac{y^2}{2}$$

$$m_y = \frac{\lambda y^2}{2L}$$

Applying Newton's II law on the segment of string of length y , from bottom,

$$T - m_y g = m_y (2g)$$

$$T = m_y (3g) = \frac{3\lambda y^2}{2L} g$$



Speed of wave on string at y ,

$$V_y = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{3\lambda y^2 g}{2L} \times \frac{L}{\lambda y}} = \sqrt{\frac{3yg}{2}}$$

$$a_y = \frac{v dv}{dy} = \frac{3g}{4}$$

$$\text{Also, } v = \frac{dy}{dt} = \sqrt{\frac{3yg}{2}}$$

$$\Rightarrow \int_0^l \frac{1}{\sqrt{y}} dy = \int_0^t \sqrt{\frac{3g}{2}} dt \Rightarrow t = \sqrt{\frac{8L}{3g}}$$

Q. If temp. of rod is decrease by $\Delta\theta$ find speed of transverse wave. [JEE Mains favourite question]



Sol. $\frac{T}{A} = y \frac{\Delta L}{L} = y \alpha \Delta\theta$

$$T = A y \alpha \Delta\theta$$

$$V_w = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{A y \alpha \Delta\theta}{\mu}} = \sqrt{\frac{T}{\rho A}}$$

$$= \sqrt{\frac{A y \alpha \Delta\theta}{\rho A}} = \checkmark$$

Power Transfer

Instantaneous power

$$P = -T \left(\frac{\partial y}{\partial x} \right) \left(\frac{\partial y}{\partial t} \right)$$

eqn of wave can be

$$y = A \sin(\omega t - kx + \phi)$$

$$\star p = TA^2 k \omega \cos^2(\omega t - kx + \phi)$$

$$\star V_w = \sqrt{\frac{T}{\mu}}$$

$$\star P = \mu A^2 \omega^2 \cos^2(\omega t - kx + \phi) V_w$$

$$\star T = \mu V_w^2 = \mu V_w \cdot V_w$$

$$\star T = \mu \frac{\omega}{k} V_w$$

$$\star \text{Average power} = \frac{1}{2} \mu A^2 \omega^2 V_w$$

$$\text{Intensity} = \frac{\text{Energy}}{\text{time} \cdot \text{Area}} = \frac{\text{Power}}{\text{Area}}$$



काम का डब्बा

★ Derivation are not important but their following result are important.

★ Power transmitted along the string

$$P = -T \left(\frac{\partial y}{\partial t} \right) \left(\frac{\partial y}{\partial x} \right)$$

★ $\frac{d(PE)}{dx} = \frac{1}{2} (\text{slope})^2$ if slope = 0, PE = 0

$$\star dK = \frac{1}{2} \mu dx (V_p)^2$$

★ For an element $d(T.E) = d(KE) + d(PE)$

★ KE, PE & TE तीनों Time के साथ variable होता है

★ For a travelling wave KE and PE of an element changes are same.

★ For $y = A \sin(\omega t - kx + \phi)$

$$\star dk = du = \frac{1}{2} \mu A^2 \omega^2 \cos^2(\omega t - kx + \phi) dx$$

$$\star d(T.E.) = \mu A^2 \omega^2 \cos^2(\omega t - kx + \phi) dx$$

$$\star \frac{d(T.E)}{dx} = \mu A^2 \omega^2 \cos^2(\omega t - kx + \phi)$$

★ जब भी कभी हमें PE देखनी होगी हम slope के बारे में सोचेंगे अगर slope zero है तो PE zero होगी चाहे Travelling wave हो चाहे standing wave है (in wave on string)

For a loop (wavelength)

$$\left. \begin{aligned} KE &= PE = \frac{1}{4} \mu A^2 \omega^2 \lambda \\ TE &= \frac{1}{2} \mu A^2 \omega^2 \lambda \end{aligned} \right\} \text{constant}$$

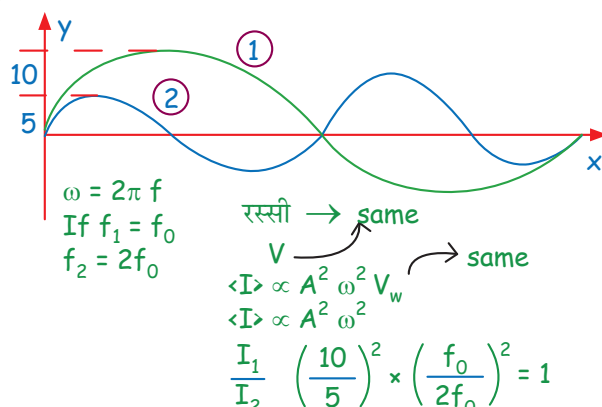
★ $P = \mu A^2 \omega^2 \cos^2(\omega t - kx + \phi) V_w$

★ $\langle I \rangle = \text{Avg Intensity} = \frac{\frac{1}{2} \mu A^2 \omega^2 V_w}{\text{Area}}$

★ $\langle I \rangle \propto A^2 \omega^2 V_w$

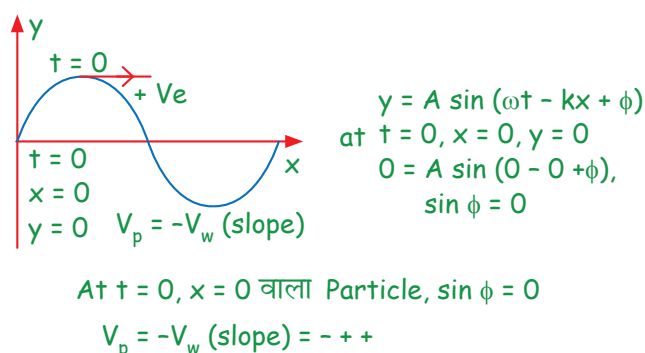
Q. Two wave 1&2 are moving in same material string in same medium with same intensity find ratio of average intensities.

Sol.

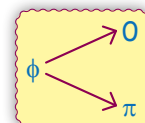


Q. Find equation of transverse wave if at $t = 0$ shape of रस्सी is given, wave is moving along +x-axis

Sol.

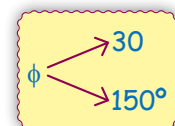
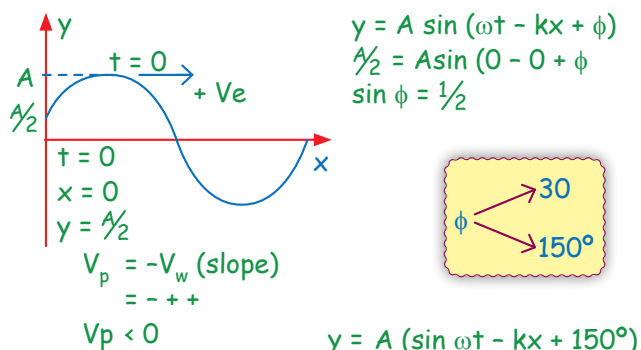


$V_p < 0$ SHM से $\Rightarrow \phi = \pi$
 $y = A \sin(\omega t - kx + \pi)$
 $y = -A \sin(\omega t - kx)$
 $y = A \sin(kx - \omega t)$



Q. Find equation of following wave moving along +ve x at $t = 0$.

Sol.



Results

Case-I

अगर $y = f(x)$ दिया है $t = 0$ पर
put $x \rightarrow x - vt$ when wave +x में जा रही है
put $x \rightarrow x + vt$ when wave -x में जाती है

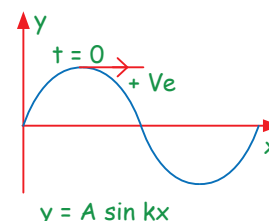
Case-II

अगर $y = f(t)$ दिया है $x = 0$ पर
put $t \rightarrow t - \frac{x}{V}$ when wave +x में जाती है
put $t \rightarrow t + \frac{x}{V}$ when wave -x में जाती है

Q. A wave is moving along -x axis with speed 10m/s such that at $t = 0$ relation between y & x given as $y = \frac{1}{x^2 + 3}$ find equation of wave

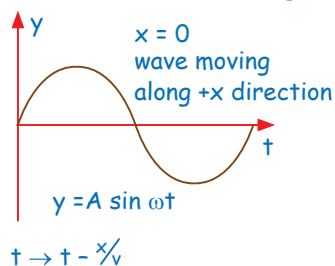
Sol. $\Rightarrow x \rightarrow x + vt = x + 10t$
 $y = \frac{1}{(x + 10t)^2 + 3}$

Q. Find equation of wave



Sol. $x \rightarrow x - vt$
 $y = A \sin k(x - vt) = A \sin(kx - kvt) = A \sin(kx - \omega t)$

Q. Find equation of wave in following.



Sol. $y = A \sin \omega \left(t - \frac{x}{v} \right)$

$$y = A \sin \left(\omega t - \frac{\omega}{v} x \right)$$

$$y = A \sin (\omega t - kx)$$

Q. At $t = 0$, a transverse wave in wire is described by function $y = \frac{6}{x^2 + 3}$ write the function (x, t) that described this pulse if wave is traveling along $+x$ axis with speed 4.5 m/s

Sol. $y = \frac{6}{(x - 4.5t)^2 + 3}$
 $x \rightarrow x - 4.5t$

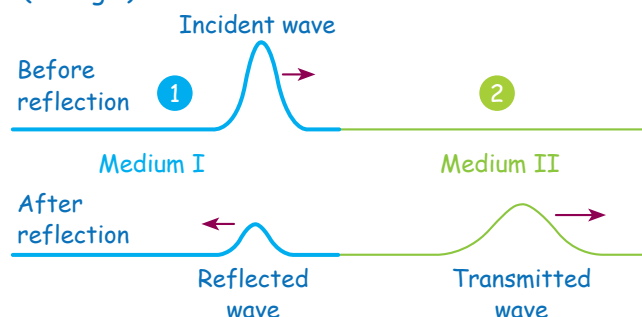
Q. A wave is traveling with speed 2 m/s along $+x$ axis such that displacement of particle at $x = 0$ is given by relation $y = \frac{2}{t^2 + 1}$ find equation of wave

Sol. $t \rightarrow t - \frac{x}{2}$
 $y = \frac{2}{(t - \frac{x}{2})^2 + 1}$

Transmission of Wave

Suppose we have two different string connected to each other as shown in fig. and a wave of eqn $y_i = A \sin (\omega t - kx)$ is moving in a string 1 of mass per unit length μ_1 (in medium I).

When pulse reaches at the junction, some part of the energy transmitted into medium II (string 2), and some part of the energy get reflected to the medium I (string 1).



After a long derivation we got result that amplitude reflected wave (A_r) and amplitude of transmitted wave (A_t) is given by

$$\star A_r = \left(\frac{V_2 - V_1}{V_2 + V_1} \right) A_i$$

$$\star A_t = \left(\frac{2V_2}{V_2 + V_1} \right) A_i$$

$V_1 \rightarrow$ Speed of wave in string 1

$V_2 \rightarrow$ Speed of wave in string 2

$A_i \rightarrow$ Amplitude of incident wave

$A_r \rightarrow$ Amplitude of reflected wave

$A_t \rightarrow$ Amplitude of transmitted wave



अबे सच्ची बता रहा हूँ इसका derivation काम का नहीं है, direct formula याद करले।

$$\star \text{ Since } A_r = \left(\frac{V_2 - V_1}{V_2 + V_1} \right) A_i$$

If $V_2 < V_1$ or $(\mu_1 < \mu_2) \Rightarrow A_r < 0$ मतलब pulse उल्टी होकर वापस आएगी $\Rightarrow$ If wave are moving from rare medium to denser medium then reflected wave will be inverted so 180° additional phase difference has to be added in reflected wave.

$\star$ Since $A_t = \left(\frac{2V_2}{V_2 + V_1} \right) A_i \Rightarrow A_t$ and A_i will have same sign so we can say transmitted pulse never inverted to incident pulse.

$\star$ जहाँ v ज्यादा $\Rightarrow$ Rare medium

$\star$ जहाँ v कम $\Rightarrow$ denser medium

Q. A progressive wave travels in a medium M_1 and enters into another medium M_2 in which its speed decreases to 25%. What is the ratio of the amplitude of the

(a) Reflected and the incident waves?

(b) Transmitted and the incident waves?

Sol. Given that, velocity in the medium refracted is 25% of that in the initial medium $v_2 = \frac{1}{4} v_1$

$$(a) \frac{A_r}{A_i} = \frac{v_2 - v_1}{v_2 + v_1} = \frac{\frac{v_2}{v_1} - 1}{\frac{v_2}{v_1} + 1} = \frac{\frac{1}{4} - 1}{\frac{1}{4} + 1} = -\frac{3}{5}$$

i.e., the required ratio is $\left| \frac{A_r}{A_i} \right| = 3 : 5$

$$(b) \frac{A_t}{A_i} = \frac{2v_2}{v_2 + v_1} = \frac{2v_2 / v_1}{\frac{v_2}{v_1} + 1} = \frac{2(1/4)}{\left(\frac{1}{4}\right) + 1} = \frac{2}{5}$$

i.e., the required ratio is $\left| \frac{A_t}{A_i} \right| = 2:5$

Q. 

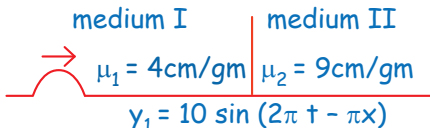
If incident wave has eqn $y_i = 10 \sin(2\pi t - \pi x)$

Find amplitude of reflected & transmitted wave

Sol. $A_t = \frac{2\sqrt{\frac{T}{\mu_2}}}{\sqrt{\frac{T}{\mu_1}} + \sqrt{\frac{T}{\mu_2}}} A_i$

$$A_t = \frac{2\sqrt{1/4}}{\sqrt{1/9} + \sqrt{1/4}} \times 10 = \frac{2/2 \times 10}{1/3 + 1/2} \text{ cm} = 12 \text{ cm}$$

$$A_r = \left(\frac{V_2 - V_1}{V_2 + V_1} \right) \times 10 = 2 \text{ cm}$$

Q. 

Find amplitude of reflected & transmitted wave.

Sol. $A_t = \frac{2V_2}{V_2 + V_1}$

$$A_i = \frac{2\sqrt{1/9}}{\sqrt{1/4} + \sqrt{1/9}} \times 10$$

$$= \frac{2/3}{1/2 + 1/3} (10) = \frac{20 \times 6}{3 \times 5} = 8$$

$$A_r = \frac{V_2 - V_1}{V_2 + V_1}$$

$$A_i = \frac{\sqrt{1/9} - \sqrt{1/4}}{\sqrt{1/9} + \sqrt{1/4}} \times 10$$

$$= \frac{1/3 - 1/2}{1/3 + 1/2} \times 10 = -2$$

★ Fixed end of a string:

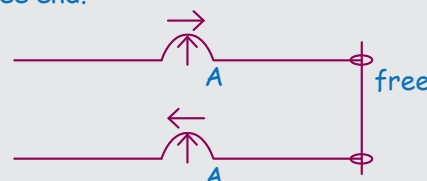
Since the fixed end is equivalent to a string of infinite linear mass density, $v_2 = \sqrt{(T/\mu)} = 0$, and we obtain,



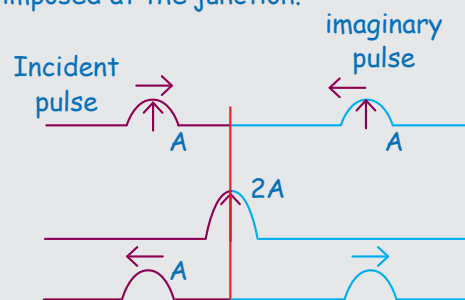
$$A_t = 0 \text{ and } A_r = -A_i$$

★ If one end of string is free

Reflected wave is not inverted here when reflected from free end.



In this case we can assume an imaginary mirror image pulse coming from another medium and they super imposed at the junction.

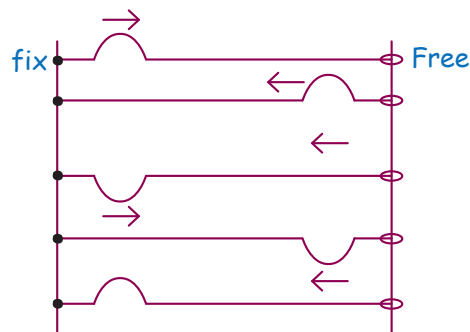


★ Both end fixed

Wave going towards right is reflected from fixed end.



Q. A pulse is going rightward. Left boundary is fixed & right boundary free. Draw reflected wave.



Standing Wave

When two waves of same amplitude & same frequency moving in opposite direction superimpose that resulting into standing wave.

$$y_1 = A \sin(kx - \omega t)$$

$$y_2 = A \sin(kx + \omega t)$$

$$y_{\text{net}} = y_1 + y_2 = A \sin(kx - \omega t) + A \sin(kx + \omega t)$$

$y_{\text{net}} = 2A \sin kx \cos \omega t$ (Equation of standing wave)

Here $A(x) = 2A \sin kx$ = Amplitude

यहाँ x की value put करने पर उस point का amplitude निकाल सकते हो which is very important for mains and adv.



Node

★ जिनका amplitude zero है, and they are at rest always

$A(x) = 0$ which are at rest always

$$2A \sin kx = 0 \text{ (in above eqn.)}$$

$$kx = n\pi$$

$$\frac{2\pi}{\lambda}x = n\pi \Rightarrow x = \frac{n\lambda}{2}$$

$$x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, 2\lambda, \dots$$

इन value पर y_{net} zero होगा

Antinode

★ जिनका amplitude maximum है, which is $2A$

$A(x) \rightarrow$ maximum (Amplitude)

$$2 \sin kx = \pm 2A$$

$$\sin kx = \pm 1$$

$$kx = 90^\circ, 270^\circ, 450^\circ$$

$$\frac{2\pi}{\lambda}x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

$$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$$



काम का डब्बा

★ Stationary waves are formed due to superposition of two identical waves travelling in opposite direction with same speed.

★ Particle having zero amplitude called node they are at rest and having max strain

★ Particle having max amplitude called antinode $\Rightarrow$ having minimum strain

★ Node & antinode are formed alternately distance between two consecutive node is $\frac{\lambda}{2}$ also distance between two consecutive antinode is $\frac{\lambda}{2}$

★ Distance between two consecutive node & AN is $\frac{\lambda}{4}$

★ All the particle in one particular segments vibrates in same plane but particle between in adjacent segment differ in phase by 180°

$$\text{Power}(p) = -T \frac{dy}{dt} \frac{dy}{dx}$$

★ Node वो point है जिसका amplitude zero है जिनका momentum, velocity, kE, displacement zero है, Antinode वो particle है जिनका Amplitude max है

$$p = -T \frac{dy}{dt} \frac{dy}{dx} \text{ node पर } \frac{dy}{dt} = 0$$

इसलिए power transfer V zero है (through node)

$$\text{OR antinode पर slope zero है } \frac{dy}{dx} = 0$$

$\Rightarrow$ Power transfer = 0

★ Power transfer through node & antinode is always zero

Q. If equation of standing wave is given by

$$y = 20 \sin \frac{\pi x}{6} \sin \frac{\pi t}{6} \text{ find following quantities.}$$

Sol. (1) Angular freq. (ω) and propagation const. (k):

$$\Rightarrow \omega = \frac{2\pi}{T} = \frac{\pi}{3}$$

$$\Rightarrow k = \frac{2\pi}{\lambda} = \frac{\pi}{6}$$

$$T = 6$$

$$\lambda = 12$$

$$\Rightarrow \text{Distance between two consecutive node} = \frac{\lambda}{2} = \frac{12}{2} = 6$$

$\Rightarrow$ Distance between consecutive AN = $\frac{\lambda}{2} = 6$

$\Rightarrow$ Distance between consecutive N & AN = $\frac{\lambda}{4} = 3$

$\Rightarrow$ Amplitude of constituent waves, when they superimposed given this standing wave $A = 10$ and $2A = 20$

(2) Find location of node & anti node

$$y = 20 \sin \frac{\pi x}{6} \sin \frac{\pi t}{3}$$

For node calculation

$$\sin \frac{\pi x}{6} = 0$$

$$\frac{\pi x}{6} = 0, \pi, 2\pi, 3\pi$$

$$x = 0, 6, 12, 18, \dots$$

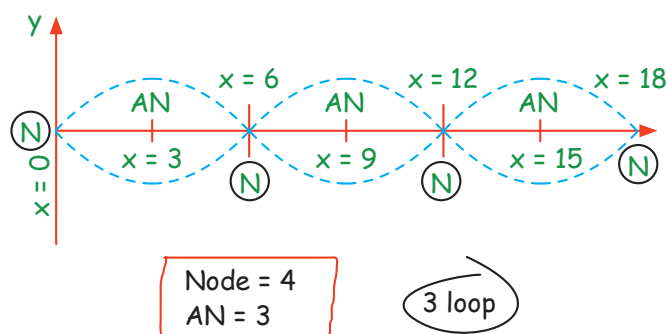
For antinode calculation

$$\sin \frac{\pi x}{6} = \pm 1$$

$$\frac{\pi x}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

$$x = 3, 9, 15, 21$$

(3) If length of string is 18 m find no. of node and anti node



Q. If eqn of standing wave is given by

$$y = 20 \sin \frac{\pi x}{6} \cos \frac{\pi t}{3}$$

(a) Find amplitude of particle at $x = 1$, $x = 2$, $x = 6$

Sol. Amplitude = $20 \sin \frac{\pi x}{6}$

$$x = 1, \text{ Amplitude} = 20 \sin \frac{\pi \times 1}{6} = 10$$

$$x = 2, \text{ Amplitude} = 20 \sin \left(\frac{\pi \times 2}{6} \right) = 10\sqrt{3}$$

$$x = 6, \text{ Amplitude} = 20 \sin \left(\frac{\pi \times 6}{6} \right) = 0 \text{ (node)}$$

(b) Find location of point which have amplitude equal to half of max amplitude

Sol. $A_{\max} = 20$

half of $A_{\max} = A = 10$

$$A = 10 = (20) \sin \frac{\pi x}{6}$$

$$\text{or } \pm = \frac{1}{2} = \sin \left(\frac{\pi x}{6} \right)$$

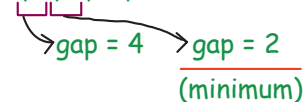
$$\frac{\pi x}{6} = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6},$$

$$\frac{\pi x}{6} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \dots$$

$$x = 1, 5, 7$$

(c) Find the minimum gap between particle whose amplitude is half of max amplitude

Sol. 1, 5, 7, 11, 13



Q. Find equation of standing wave also find location of Node & AN formed by two waves y_1 and y_2

$$y_1 = A \sin (\omega t - kx)$$

$$y_2 = A \sin (\omega t + kx)$$

Find position of node & anti nodes

कुछ नहीं करना है बस $y_1 + y_2$ करना है... Try कर

Sol. $y_{\text{net}} = y_1 + y_2 = A \sin \omega t \cos kx - A \cos \omega t \sin kx + A \sin \omega t \cos kx + A \cos \omega t \sin kx$

$$y_{\text{net}} = 2A \cos kx \sin \omega t$$

For Node $\Rightarrow \cos kx = 0$

$$\frac{2\pi}{\lambda} x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

$$x = \frac{\lambda}{4}, \frac{3\lambda}{4} \text{ (Position of nodes)}$$

For Antinode $\Rightarrow$

$$\cos kx = \pm 1$$

$$kx = n\pi$$

$$\frac{2\pi}{\lambda} x = n\pi$$

$$x = n\frac{\lambda}{2}$$

$$x = 0, \frac{\lambda}{2}, \lambda \text{ (Position of antinodes)}$$

Q. If $y = 10 \sin \frac{\pi x}{12} \cos \frac{\pi t}{6}$ describes standing wave, answer the following parts.

Sol. 1. Simplifying the wave equation, $\lambda = 24$ m

$$\omega = \frac{2\pi}{T} = \frac{\pi}{6}$$

$$T = 12 \text{ sec}$$

$$K = \frac{\pi}{12} = \frac{2\pi}{\lambda} = \frac{\pi}{12}$$

2. Distance between two consecutive node $\frac{\lambda}{2} = 12$

3. Distance between two consecutive AN $\frac{\lambda}{2} = 12$

4. Distance between two consecutive N & AN $= \frac{\lambda}{4} = 6$

5. Location of node

$$\sin \frac{\pi x}{12} = 0$$

$$\frac{\pi x}{12} = n\pi$$

$$x = n \times 12$$

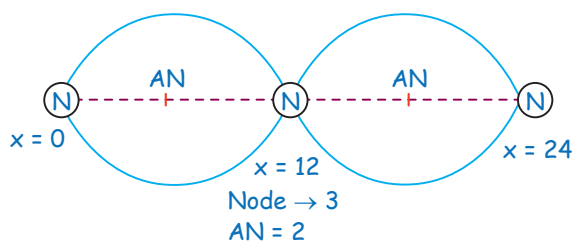
$$x = 0, 12, 24$$

6. Location of AN

$$\sin \frac{\pi x}{12} = \pm 1$$

$$\frac{\pi x}{12} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

7. If length of string is 24 meter find no. of node & antinode.



Q. The vibrations of a string fixed at both ends are described by the equation

$$y = (5.00 \text{ mm}) \sin [(\pi/2 \text{ cm}^{-1})x] \sin [(314 \text{ rads}^{-1})t]$$

- What is the maximum displacement of the particle at $x = 5.66$ cm?
- What are the wavelengths and the wave speeds of the two transverse waves that combine to give the above vibration?
- What is the velocity of the particle at $x = \frac{17}{3}$ cm at time $t = 2.00$ s?

(iv) If the length of the string is 10.0 cm, locate the nodes and the anti-nodes. How many loops are formed in the vibration?

Sol. The transverse velocity of particle of string is

$u = \frac{\partial y}{\partial t}$. The wave velocity is $v = f\lambda$. Comparing wave equation with $y = A \sin kx \sin \omega t$, we get the amplitude (A) and angular frequency (ω) of the wave.

(i) The amplitude of the vibration of the particle at position x is

$$A = (5.00 \text{ mm}) \sin [(\pi/2 \text{ cm}^{-1})x]$$

Putting $x = \frac{17}{3}$ cm, and re-writing the expression

$$A = (5.00 \text{ mm}) \sin \left[\frac{\pi}{2} \times \frac{17}{3} \right]$$

$$\text{or} = \left| (5.00 \text{ mm}) \sin \left(2\pi + \frac{5\pi}{6} \right) \right|$$

$$= \left| (5.00 \text{ mm}) \sin \left(\frac{\pi}{6} \right) \right| = 2.50 \text{ mm}$$

(ii) From the given equation, the wave number

$k = \frac{\pi}{2} \text{ cm}^{-1}$ and the angular frequency $\omega = 314 \text{ rad.s}^{-1}$. Thus, the wavelength is

$$\lambda = \frac{2\pi}{k} = \frac{2 \times 3.14}{1.57 \text{ cm}^{-1}} = 4.00 \text{ cm}$$

and frequency is

$$f = \frac{\omega}{2\pi} = \frac{314 \text{ rads}^{-1}}{2 \times 3.14} = 50 \text{ s}^{-1}$$

The wave speed is

$$v = f\lambda = (50 \text{ s}^{-1})(4.00 \text{ cm}) = 2.00 \text{ ms}^{-1}$$

(iii) The velocity of the particle at position x at time t is given by

$$u = \frac{\partial y}{\partial t} = (5.00 \text{ mm}) \sin \left[(1.57 \text{ cm}^{-1})x \right]$$

$$\left[314 \text{ s}^{-1} \times \cos(314 \text{ s}^{-1})t \right]$$

$$\frac{\partial y}{\partial t} = (5.00 \text{ mm}) \sin \left(\frac{\pi}{2} \text{ cm}^{-1}x \right) \{ 200\pi \cos(200\pi t) \}$$

Putting $x = \frac{17}{3}$ cm and $t = 2$ s, we have

$$\frac{\partial y}{\partial t} = (5 \text{ mm} \times 200\pi) \sin\left(\frac{\pi}{2} \text{ cm}^{-1} \times \frac{17}{3} \text{ cm}\right) \cos(400\pi t)$$

$$= (5 \times 10^{-3} \text{ m} \times 200 \times 3.14 \text{ s}^{-1}) \sin\left(2\pi + \frac{5\pi}{6}\right) \times 1$$

$$= (3.14 \text{ ms}^{-1}) \sin \frac{\pi}{6}$$

$$\text{or } v = 1.57 \text{ m/s} = 157 \text{ cms}^{-1}$$

(iv) The nodes occur where the amplitude is zero, i.e.,

$$\sin\left(\frac{\pi}{2} \text{ cm}^{-1}\right) x = 0 \text{ or } \left(\frac{\pi}{2} \text{ cm}^{-1}\right) x = n\pi$$

where n is an integer. Thus, $x = 2n \text{ cm}$.

($n = 0, 1, 2 \dots$)

The nodes, therefore, occur at $x = 0 \text{ cm}, 2 \text{ cm}, 4 \text{ cm}, 6 \text{ cm}, 8 \text{ cm}$ and 10 cm . Anti-nodes occur in between the nodes, i.e., at $x = 1 \text{ cm}, 3 \text{ cm}, 5 \text{ cm}, 7 \text{ cm}$ and 9 cm . The string vibrates in 5 loops.

Different form of equation of standing wave can be

$$y = 2A \sin kx \cos \omega t$$

$$y = 2A \sin kx \sin \omega t$$

$$y = 2A \cos kx \cos \omega t$$

$$y = 2A \cos kx \sin \omega t \dots \dots \dots \text{etc etc.}$$

★ Energy in standing wave (wave on string)

$$\text{For one loop} \Rightarrow T.E = \frac{1}{8} \mu A^2 \omega^2 \lambda$$

$$\text{For half loop} \Rightarrow T.E = \frac{1}{16} \mu A^2 \omega^2 \lambda$$

भाई इसका derivation बिल्कुल important नहीं है only formula important है इसे अच्छे से याद करलो

STATIONARY WAVES IN STRINGS

Case I-Both Ends of String are Fixed

Fundamental Mode (0^{th} overtone, first harmonic)

The string vibrates in one loop in which the ends are the nodes and the centre is the antinode. This mode of vibration is known as the fundamental mode and frequency of vibration is known as the fundamental frequency or first harmonic.

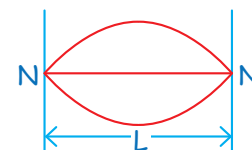
$$\frac{\lambda}{2} = L$$

$$\lambda = 2L$$

$$f = \frac{V}{\lambda} = \frac{V}{2L}$$

$$V = \sqrt{\frac{T}{\mu}}$$

$$f_0 = \frac{V}{2L}$$

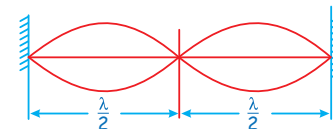


Second harmonic, 1st overtone

$$L = \lambda$$

$$f = \frac{V}{\lambda} = \frac{V}{L} = \frac{2V}{2L}$$

$$f = 2 \left(\frac{V}{2L} \right)$$

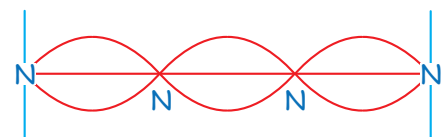


Third harmonic, 2nd overtone

$$\frac{3\lambda}{2} = L$$

$$\lambda = \frac{2L}{3}$$

$$f = \frac{V}{\lambda} = \frac{3V}{2L} = 3 \left(\frac{V}{2L} \right) \text{ (3 loop)}$$



When string fixed at both end

★ Fundamental node $f_0 = \frac{V}{2L}$ $V = \sqrt{\frac{T}{\mu}}$

★ n^{th} harmonic = $f_n = n \frac{V}{2L}$

(Here there will be n loop, $n+1$ node, n antinode)

★ $(n-1)$ overtone = n^{th} harmonic

★ Fifth harmonic = five loop

$$= \text{node } 6$$

$$= AN = 5$$

★ 7th harmonic = (6th overtone)

$$f_7 = \frac{7V}{2L}$$

★ 5th overtone = (6th harmonic)

$$f_6 = \frac{6V}{2L}$$

Q. What are the three lowest freq for standing wave in wire of length 10m having mass 100gm which is stretched under the tension of 1600N string is fixed at both end $L = 10 \text{ m}$

Sol. $V_w = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{1600}{0.1/10}} = \sqrt{160000} = 400 \text{ m/s}$

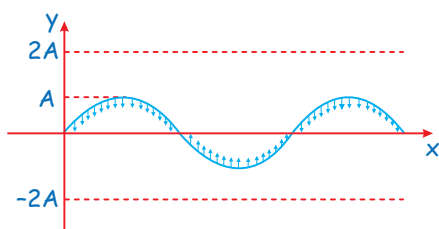
F.M = Fundamental mode 1st

$$f_0 = \frac{v}{2L} = \frac{400}{2 \times 100} = 20$$

$$f_2 = 40$$

$$f_3 = 60$$

Q. Figure shows a standing wave pattern on a string at $t = 0$. Find out the equation of the standing wave if the amplitude of antinode is $2A$.



Sol. Let us assume that the equation of the given standing wave is:

$$y = 2A \sin(kx + \phi_1) \sin(\omega t + \phi_2)$$

$$\text{At } x = 0, y = 0 \text{ (node)}$$

$$y = A \sin(\phi_1) \sin(\omega t + \phi_2) = 0 \Rightarrow \phi_1 = 0$$

$$\text{So, } y = 2A (\sin kx) \sin(\omega t + \phi_2)$$

$$\text{At } t = 0, x = \frac{\lambda}{4}, y = A$$

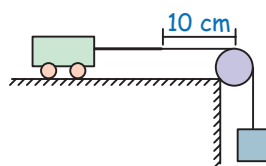
$$\Rightarrow 2A \sin \phi_2 \Rightarrow \sin(\phi_2) = \frac{1}{2} \Rightarrow \phi_2 = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

As seen from figure velocity is negative

$$\text{so, } \phi_2 = \frac{5\pi}{6}$$

$$y = 2A \sin(kx) \sin \left(\omega t + \frac{5\pi}{6} \right)$$

Q. A heavy string is tied at one end to a movable support and to a light thread at the other end as shown in figure. The thread goes over a fixed pulley and supports a weight to produce a tension. The lowest frequency with which the heavy string resonates is 120 Hz. If the movable support is pushed to the right by 10 cm so that the joint is placed on the pulley, then the minimum frequency at which the heavy string can resonate is



Sol. Initially the condition at the two ends of rod are different one is node (fixed) other is anti-node (free).

Hence fundamental or lowest frequency is $\frac{v}{4L}$ where L is length of heavy string.

$$\Rightarrow f_1 = \frac{v}{4L} = 120 \quad \dots (i)$$

Now, in second case, conditions at the two ends of string become similar (both ends are nodes (fixed)),

$$\text{So, minimum frequency } f_2 = \frac{v}{2L}$$

According to question,

$$\therefore f_1 \lambda_1 = f_2 \lambda_2$$

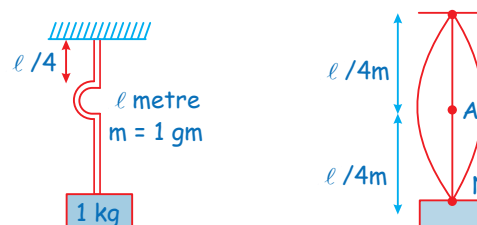
$$120 \times 4L = f_2 \times 2L \Rightarrow f_2 = 240 \text{ Hz}$$

Q. Imagine a very thin uniform wire of length ℓ meter and total mass 1 gram. Fix one end of the wire in the ceiling, hang a large mass of 1 kg from the other end, and then pluck the wire $\frac{\ell}{4}$ m from the top. Assume that as the wire vibrates the large mass is not moving. The lowest frequency obtained is 100 Hz. What is the length of the wire (in meter)? (You may ignore the effect of gravity on the mass of the wire) [Take $g = 10 \text{ m/s}^2$]

$$\text{Sol. } \mu = \frac{10^{-3} \text{ kg}}{\ell \text{ (metre)}} = \frac{10^{-3}}{\ell} \text{ kg/m}$$

$$T \approx mg = (1)(10) = 10 \text{ N}$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{10}{10^{-3}/\ell}} = \sqrt{10^4 \ell} = 10^2 \sqrt{\ell}$$



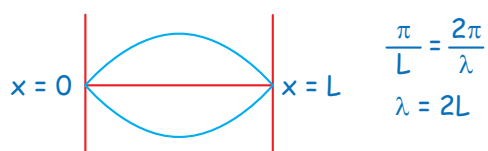
$$\frac{\ell}{4} = \frac{\lambda}{4} \Rightarrow \lambda = \ell \text{ m}$$

$$v = f \lambda \Rightarrow 10^2 \sqrt{\ell} = 100 \lambda$$

$$\therefore \ell = 1 \text{ m}$$

Q. Ends of stretched wire of length L are fixed at $x = 0$ & $x = L$ in one experiment $y_1 = A \sin \frac{\pi x}{L} \sin \omega t$ & total energy is E_1 In another experiment $y_2 = A \sin 2 \left(\frac{\pi x}{L} \right) \sin (2 \omega t)$ & total energy is E_2 find $\frac{E_1}{E_2} = ?$

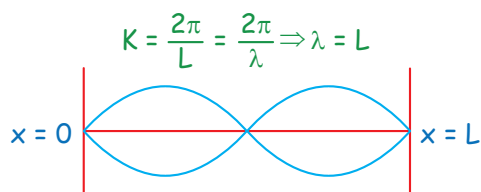
Sol. 1. First experiment



$$TE = \frac{1}{8} \mu A^2 \omega^2 \lambda$$

$$E_1 = \frac{1}{8} \mu A^2 \omega^2 \lambda = \frac{1}{8} \mu A^2 \omega^2 \lambda (2L)$$

2. 2nd experiment



$$E_2 = (\frac{1}{8} \mu A^2 \omega^2 \lambda) \times 2 = \frac{1}{8} \mu A^2 (2\omega^2) (L) \times 2$$

$$\frac{E_1}{E_2} = \frac{\omega^2}{L \times 2} = \frac{1}{4}$$

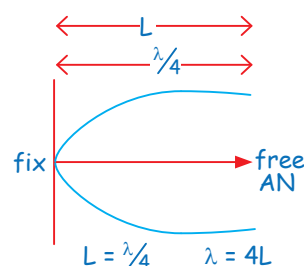
Case II-One End of String is Free

Fundamental frequency (F.m)/1st harmonic/zero overtone

$$v = f\lambda$$

$$f = \frac{v}{\lambda} = \frac{v}{4L}$$

$$f_0 = \frac{v}{4L}$$

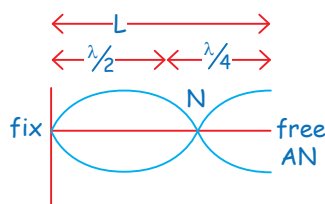


3rd harmonic/1st overtone

$$f = \frac{v}{\lambda} = \frac{3v}{4L}$$

$$f = 3\left(\frac{v}{4L}\right)$$

$$L = \frac{3\lambda}{4} \Rightarrow \lambda = \frac{4L}{3}$$

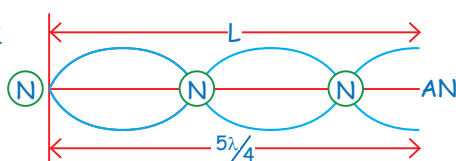


5th harmonic/2nd overtone

$$L = \frac{5\lambda}{4} \Rightarrow \lambda = \frac{4L}{5}$$

$$f = \frac{v}{\lambda} = \frac{5v}{4L}$$

$$f = 5\left(\frac{v}{4L}\right)$$



One end of string is free

$$F.M = f_0 = \frac{v}{4L}$$

$$(2n+1)^{\text{th}} \text{ harmonic} \Rightarrow f = (2n+1) \frac{v}{4L} \text{ (n}^{\text{th}} \text{ overtone)}$$

$$5^{\text{th}} \text{ overtone} = 11 \text{ harmonic (catch it)}$$



काम का डब्बा

★ देख यार जब हम class में ये दोनों case पढ़ते हैं तो बहुत आसान लगता है (string fix at both end and string free at one end) लेकिन मैंने काफी बार देखा है exam के वक्त कुछ बच्चे दोनों result में confuse हो जाते हैं और question गलत कर देते हैं so मैं नीचे एक बार फिर से result and table बनाकर data लिख रहा हूँ इसे catch करने की कोशिश करना और exam से पहले जरूर देख कर जाना।

String fixed at both end

$$\Rightarrow FM = f_0 = \frac{v}{2L}$$

$$v = \sqrt{\frac{T}{\mu}}$$

$$n^{\text{th}} \text{ harmonic} \Rightarrow f_n = \frac{nv}{2L}$$

$$n^{\text{th}} \text{ harmonic} = (n-1) \text{ overtone}$$

String fixed at one end & free at one end

$$f_0 = \frac{v}{4L}$$

$$(2n+1)^{\text{th}} \text{ harmonic} \Rightarrow n^{\text{th}} \text{ overtone}$$

$$\Rightarrow f = (2n+1) \frac{v}{4L} \Rightarrow (2n+1) \text{ harmonic}$$

	Both end fix	One end fix
7 th harmonic	$\frac{7v}{2L}$	$\frac{7v}{4L}$
11 th harmonic	$\frac{11v}{2L}$	$\frac{11v}{4L}$
4 th overtone	$\frac{5v}{2L}$ = 5 th harmonic	9 th harmonic $\frac{9v}{4L}$
6 th harmonic	$\frac{6v}{2L}$	नहीं बनेगा
10 th overtone	$\frac{11v}{2L}$	$\frac{21v}{4L}$

Q. A string fix at both end vibrating in fundamental mode of frequency 50Hz now one end is kept free find freq if it vibrates in $f_0 = 50 = v/2L$ (f.m)

(a) 5th harmonic = $f_5 = \frac{5V}{4L} = \frac{5}{4} \times 100 = 125$

(b) 5th overtone = 11th harmonic
 $= 11 \times \frac{V}{4L} = \frac{11}{4} \times 100 = \frac{1100}{4}$

Q. Two similar wires on a sonometer are tuned to unison. One wire is 25 cm long and is stretched by 100 gram-weight. Find the length of the other wire which is stretched by 400 gram-weight.

Sol. Here, $\ell = 0.25$ m, $m = 100$ g wt = 0.1 kg

So, $f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$

$f_A = \frac{1}{2\ell_A} \sqrt{\frac{T_A}{\mu}} = \frac{1}{2(0.25)} \sqrt{\frac{0.1g}{\mu}}$

$f_B = \frac{1}{2\ell_B} \sqrt{\frac{T_B}{\mu}} = \frac{1}{2\ell_B} \sqrt{\frac{0.4g}{\mu}}$

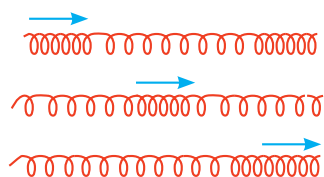
$\therefore \frac{f_A}{f_B} = \frac{2\ell_B}{0.5} \sqrt{\frac{1}{4}}$

$f_A = f_B = f$ = frequency of tuning fork

So, $1 = \frac{2\ell_B}{0.5(2)}$

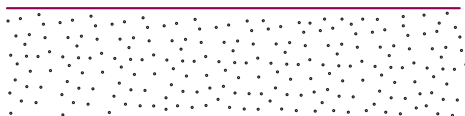
$\Rightarrow \ell_B = 0.5$ m = 50 cm

LONGITUDINAL WAVE

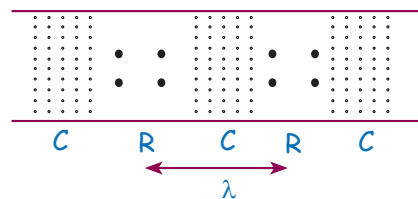


SOUND WAVE

It is Longitudinal wave particle vibrates in the direction of wave motion (parallel/anti parallel). As sound travel through air, the element of air vibrates along the direction of motion of the wave.



when sound wave travel



Equation of Sound Wave

$$S = S_0 \sin(\omega t - kx + \phi)$$

S = Displacement of particle (at an instant)

S_0 = Amplitude

$$V_w = \frac{\omega}{k}$$

ω, k sign opposite $\Rightarrow +x$ direction $\Rightarrow$

ω, k sign same $\Rightarrow -x$ direction

$$\frac{\partial S}{\partial t} = v_p \text{ (velocity)}$$

$$\frac{\partial^2 S}{\partial t^2} = a_p \text{ (acceleration)}$$

$$V_p = -V_w \frac{ds}{dw}$$

Variation of excess pressure in gas due to propagation of longitudinal wave

Let s is displacement of particle at, x and $s + ds$ is displacement of particle at $x + dx$
 change in volume of elements = ΔV

V_i = Initial volⁿ = Adx

ΔV = change in volⁿ = Ads

$$\frac{\Delta V}{V_i} = \frac{Ads}{Adx} = \frac{\partial s}{\partial x}$$

$B = \frac{-\Delta P}{\Delta V/V}$ excess pressure

$$\Delta P = -B \frac{\Delta v}{V} = -B \frac{\partial s}{\partial x}$$

ΔP - excess pressure

Let eqⁿ of sound wave be

$S = S_0 \sin(\omega t - kx + \phi) \rightarrow$ displacement eqn

& $\Delta P = -B \frac{\partial s}{\partial x}$

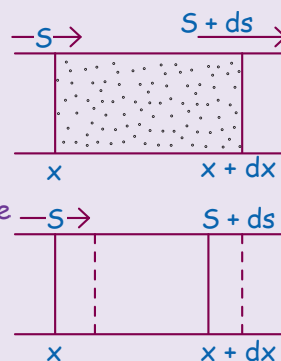
so $\Delta P = +BS_0 K \cos(\omega t - kx + \phi) \rightarrow$ pressure wave

$\Delta P = \Delta P_0 \cos(\omega t - kx + \phi)$

$\Delta P \rightarrow$ excess pressure

$\Delta P_0 \rightarrow$ excess pressure amplitude

Displacement and pressure eqn have $\frac{\pi}{2}$ phase difference



अबे डर क्यों रहा है ऊपर वाले box में derivation है which you can skip अब इसमें काम की बात सुन

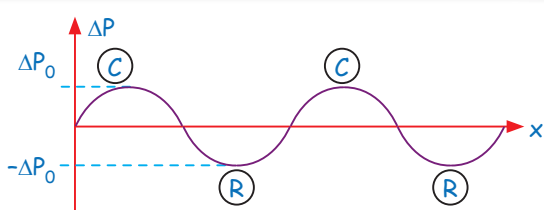
Sound wave की eqn हम दो प्रकार से represent करते हैं एक displacement से और दूसरा pressure variation से अगर displacement वाली eqn given है बोले तो $S = f(x, t)$ इससे pressure वाली eqn निकालने के लिए apply $\Delta P = -B \frac{\partial S}{\partial x}$

★ $\Delta P_0 = BS_0K$ = excess pressure amplitude

★ $P = P_0 + \Delta P_0$ $P_0 \rightarrow$ Atmospheric pressure

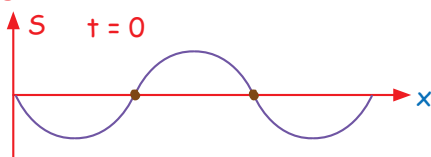
★ At compression point $P = P_0 + BS_0K$

★ At rarefaction point $P = P_0 - BS_0K$



C-Compression and R-Rarefaction

Q. Find (C) & (R) Point for S-x graph as shown in fig.

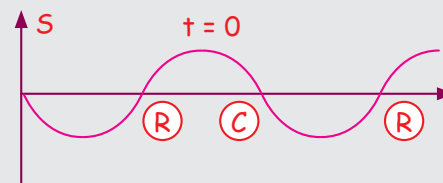
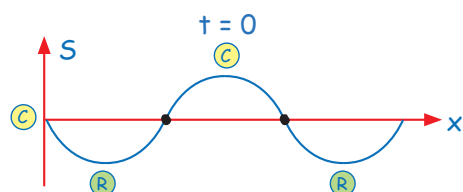
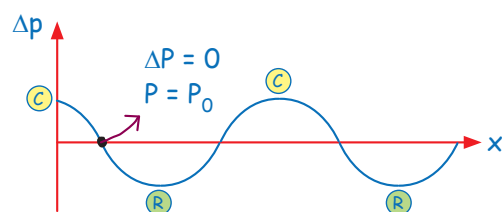
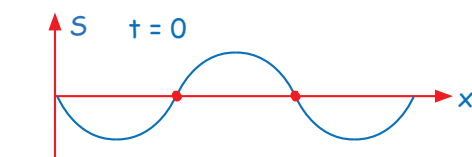


Sol. eqⁿ of graph is $S = -S_0 \sin kx$

$$\Delta p = -B \frac{ds}{dx}$$

$$\Delta p = +BS_0 k \cos kx$$

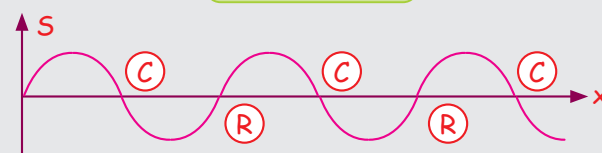
$$\Delta p = \Delta P_0 \cos kx$$



In S-x graph

Slope +Ve $\Rightarrow$ (R)

Slope -Ve $\Rightarrow$ (C)



Q. A sound wave of wavelength 40 cm travel in air. If the difference between max and minimum pressure at a point is $4.0 \times 10^{-3} \text{ N/m}^2$, find the amplitude of vibration of particle of medium.

(Bulk modulus of air = $1.4 \times 10^5 \text{ N/m}^2$)

Sol. $\lambda = 0.40$

$$K = \frac{2\pi}{\lambda} = \frac{2\pi}{0.40} = 5\pi$$

$$4 \times 10^{-3} = (P_0 + BS_0K) - (P_0 - BS_0K)$$

$$4 \times 10^{-3} = 2BS_0K$$

$$S_0 = \frac{4 \times 10^{-3}}{2 \cdot B \cdot K} = \frac{4 \times 10^{-3}}{2 \times 1.4 \times 10^5 \times 5 \times \pi}$$

Q. Equation of sound wave is given by $P = 0.2 \sin (3000t - 9x)$ where all variable are in SI unit. Answer following parts

1. Find λ , frequency, speed of sound

Sol. $\omega = 2000 = 2\pi f$

$$f = \frac{2000}{2\pi}$$

$$k = 9 = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{2\pi}{\lambda}, V_w = \frac{\omega}{K} = \frac{3000}{9} = \frac{1000}{3}$$

2. If the equilibrium pressure of air is $1.0 \times 10^5 \text{ N/m}^2$ what are the max & min pressure at a point as wave passes through the point

Sol. $P_0 = 1.0 \times 10^5$

$$S_0 = \frac{\Delta P_0}{BK}$$

$$P_{\max} = P_0 + 0.2$$

जैसे हमने wave on string मे formula देखे थे same उसी pattern पर यहाँ भी formula है मै नीचे दोनो को एक साथ लिख रहा हूँ ताकि कोई confusion ना रहे इन्हे अच्छे से समझ लेना।

Wave on String

$$\star y = A \sin(\omega t - kx + \phi)$$

$$\star V_p = \frac{\partial y}{\partial t}$$

$$\star V_p = -V_w \frac{\partial y}{\partial x}$$

$$\star a_p = \frac{\partial^2 y}{\partial t^2}$$

$$\star V_w = \frac{\omega}{K} = \sqrt{\frac{T}{\mu}}$$



Sound wave

$$\star S = S_0 \sin(\omega t - kx + \phi)$$

$$\star V_p = \frac{\partial S}{\partial t}$$

$$\star V_p = -V_w \frac{\partial S}{\partial x}$$

$$\star a_p = \frac{\partial^2 S}{\partial t^2}$$

$$\star V_{\text{sound in gas}} = \sqrt{\frac{\gamma RT}{m}} = \sqrt{\frac{\gamma P}{\rho}}$$

But actual value of velocity sound in air is 332 m/s so this result has approx 15% error..... but at that time



Then Laplace corrected this error

Laplace Correction

According to Laplace Propagation of sound is Adiabatic process in air. In adiabatic process $B = \gamma P$ velocity of sound is given by

$$V_w = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{m}} \quad \text{gamma } \gamma = \frac{C_p}{C_v}$$

For air $\gamma = 1.4$ and taking the value of P and ρ as before, we get $v = \sqrt{1.4} \times 280 = 330 \text{ m/s}$ which is quite close to the experimental value. So now velocity of sound will be

$$V_w = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{m}}$$

ओ भाई यहाँ temperature kelvin मे है गलती मत कर देना

VELOCITY OF SOUND IN A MEDIUM

$$V_w = \sqrt{\frac{E}{\rho}} \quad (\text{Longitudinal wave})$$

$E \Rightarrow$ Elastic coefficient of medium

$\rho \Rightarrow$ density

1. For solid $E = Y$ (young's modulus)

$$V_w = \sqrt{\frac{Y}{\rho}}$$

2. Fluid/gas $E = B =$ bulk modulus

$$V_w = \sqrt{\frac{B}{\rho}}$$

Velocity of Sound in Air

According to Newton sound propagates by isothermal process

In isothermal process, bulk modulus is equal to pressure

$$B = P$$

$$V_w = \sqrt{\frac{P}{\rho}} = \sqrt{\frac{RT}{m}} \approx 280 \text{ m/s} \quad m \rightarrow \text{molar mass}$$

Q. If velocity of sound in oxygen gas is V_0 find velocity of sound in nitrogen gas at same temp.

Sol. $V = \sqrt{\frac{\gamma RT}{m}} \quad \frac{\gamma}{T} \Rightarrow \text{same}$

$$\frac{V_1}{V_2} = \sqrt{\frac{m_2}{m_1}}$$

$$\frac{V_{N_2}}{V_0} = \sqrt{\frac{32}{28}}$$

Q. Find velocity of sound in oxygen at 20°C is V_0 . Find velocity of sound in He at 40°C .

Sol. Velocity in oxygen $\Rightarrow V_0 = V_1 = \sqrt{\frac{\gamma_1 R(273 + 20)}{m_1}}$

$$\text{Velocity in He} \Rightarrow V_2 = \sqrt{\frac{\gamma_2 R(273 + 40)}{m_2}}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{\gamma_1}{\gamma_2} \frac{293}{313} \frac{m_2}{m_1}}$$

$$O_2 = \gamma_{\text{dia}} = 7/5 = \gamma_1$$

$$He = \gamma_{\text{mono}} = 5/3 = \gamma_2$$

Q. If pressure of gas is increased to four times at const temp velocity of sound excepted will be

Sol. $V_w = \sqrt{\frac{\gamma RT}{m}} \Rightarrow V_w$ will be Same

Since temperature is constant so V_w will be constant.

अब कुछ बच्चे सोचेंगे सर $V_w = \sqrt{\frac{\gamma P}{\rho}}$ लगाएँ तो $V_w \sqrt{2}$ times हो जा रहा है..... अरे भाई नीचे ρ भी तो है भुल गया क्या $PM = \rho RT$



Factors Affecting Velocity of Sound in Gaseous Medium

- (i) Effect of pressure at constant temperature:
No effect,
- (ii) Effect of temperature:

$$v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{T}$$

★ For a given gas:

$$\frac{v_1}{v_2} = \sqrt{\frac{273 + t_1}{273 + t_2}}$$

where t_1 and t_2 are in $^{\circ}\text{C}$

★ If velocity of sound at 0°C in gas is v_0 , then at $t^{\circ}\text{C}$

$$v_t = v_0 + 0.61 t \text{ m/sec}$$

where $v_0 = 331 \text{ m/sec}$.

★ On increasing temperature by 1°C velocity of sound increases by 0.61 m/sec

★ At same temperature velocity in different gases

$$\frac{v_1}{v_2} = \sqrt{\frac{\gamma_1 M_2}{\gamma_2 M_1}}$$

- (iii) Effect of Humidity: On increasing humidity, velocity of sound increases because density decreases with humidity increase. Dry air has more density than humid air.

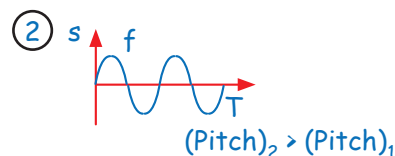
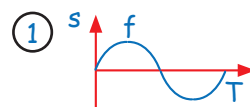
$\therefore$ Velocity of sound in dry air is less than in humid air.

Characteristic of Sound

Pitch

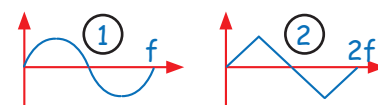
★ depend on freq

★ $\text{freq} \uparrow \text{Pitch} \uparrow$



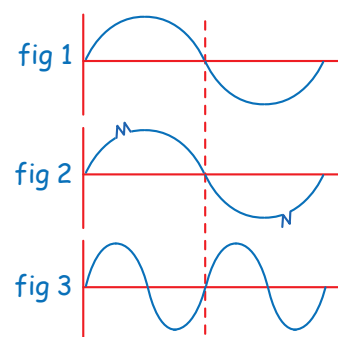
Quality

The sound from different musical instruments have different quality. It depends on wave form



In above two graph we can say
freq $\rightarrow$ same: Pitch $\rightarrow$ same
quality $\Rightarrow$ different

Q. Analysis following graph on basis of frequency of quality



Sol. Fig 1 and 3 have same quality but diff freq
Fig 1 and 2 have diff quality but same freq

Loudness $\rightarrow$ Depend on Intensity

$I \rightarrow$ Energy per unit area per unit time is called intensity.

Loudness depends on intensity of sound, loudness of sound is measured in decibel (dB)

$$\text{Loudness} = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

$I_0 =$ min Intensity that can be heard by normal person
 $= 10^{-12} \text{ w/m}^2$

Q. If $I = 10^{-10}$ watt/m². Calculate Loudness.

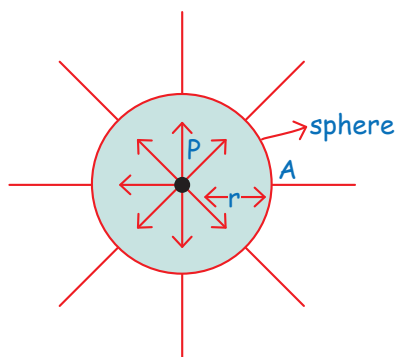
Sol. $L = 10 \log \left(\frac{10^{-10}}{10^{-12}} \right) = 20 \text{ dB}$

Intensity: Energy per unit area per unit time or power per unit area

1. point source: Intensity at 'A'

$$I = \frac{P}{4\pi r^2}$$

$$I \propto \frac{1}{r^2} \text{ point source}$$



Q. Find ratio of intensity at point A & B due to point source.



Sol. $\Rightarrow I_A = \frac{P}{4\pi r^2}$

$$I_B = \frac{P}{4\pi (2r)^2}$$

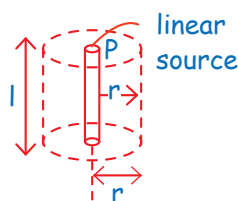
$$\frac{I_A}{I_B} = \frac{(2r)^2}{(r)^2}$$

2. Linear source:

Intensity at distance r

$$I_A = \frac{P}{2\pi r l}$$

$$I \propto \frac{1}{r} \text{ linear source}$$



3. Parallel beam:

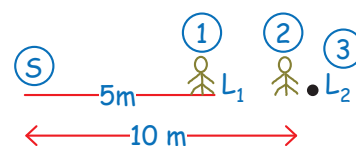
Intensity is independent of distance

$$I \propto r^0$$

$$I \rightarrow \text{const}$$

Q. Consider a point sources emitting sound wave, find difference in decibel scale reading (find difference in loudness) between two points at a distance 5m & 10m from source respectively.

Sol. $L_1 = 10 \log \frac{I_1}{I_0}$
 $L_2 = 10 \log \frac{I_2}{I_0}$



$$L_1 - L_2 = 10 \left(\log \frac{I_1}{I_0} - \log \frac{I_2}{I_0} \right) \quad [I \propto 1/r^2]$$

$$= 10 \log \frac{I_1}{I_2} = 10 \log \left(\frac{r_2}{r_1} \right)^2 \text{ and } \frac{I_1}{I_2} = \left(\frac{r_2}{r_1} \right)^2$$

$$= 20 \log \left(\frac{10}{5} \right) = 20 \log 2$$

Q. A certain sound level is increased by 20 dB by what multiple its intensity (source is point source)

Sol. Suppose sound level increases from L_1 to L_2 & intensity from I_i to I_f

$$L_2 - L_1 = 20$$

$$10 \log \frac{I_f}{I_0} - \log \frac{I_i}{I_0} = 20$$

$$10 \log \frac{I_f}{I_i} = 20$$

$$\log \frac{I_f}{I_i} = 2$$

$$\frac{I_f}{I_i} = 10^2 = 100$$

$$I_f = 100I_i$$

Q. The loudness level at a distance R from a long linear source is found to be 40dB. At this point, the amplitude of oscillation of air molecules is 0.01cm. Then find the loudness level and amplitude at a point at distance $10R$ from the source.

Sol. For a linear source we know

$$\text{Intensity } (I) \propto \frac{1}{R} \text{ (R: distance from source)}$$

$$\text{and amplitude } (A) \propto \frac{1}{\sqrt{R}}$$

$$\text{Loudness } (L) = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

Where I is intensity at ' R ' and I_0 barely audible intensity

Let the intensities at $r_2 = 10R$ and $r_1 = R$ be I_2 & I_1 respectively.

Hence, loudness at $r_2 = 10R$, $L_2 = 10 \log_{10} \left(\frac{I_2}{I_0} \right)$

and loudness at $r_1 = R$, $L_1 = 10 \log_{10} \left(\frac{I_1}{I_0} \right)$

$$\Rightarrow L_2 - L_1 = 10 \log_{10} \left(\frac{I_2}{I_1} \right)$$

$$\text{or } L_2 - 40 \text{ dB} = 10 \log_{10} \left(\frac{1/10R}{1/R} \right) \quad \left[\because I \propto \frac{1}{R} \right]$$

$$L_2 - 40 \text{ dB} = -10 \text{ dB}$$

$$\Rightarrow L_2 = 30 \text{ dB}$$

$\therefore$ Amplitude at $r_1 = R$

Let amplitude at $r_2 = 10R$ be A_2

$$\Rightarrow A_2 = A_1 \left(\frac{\sqrt{R}}{10R} \right) \Rightarrow A_2 = \frac{0.01}{\sqrt{10}} \text{ cm} = 10\sqrt{10} \mu\text{m}$$

Intensity in Sound Wave

Suppose eqn of sound wave be

$$S = S_0 \sin(\omega t - kx)$$

Pressure will vary as

$$\Delta p = BS_0 k \cos(\omega t - kx)$$

$$\text{Power} = P = \vec{F} \cdot \vec{V} = F \cdot \frac{ds}{dt} \cos O$$

$$= \Delta p A \frac{ds}{dt} = BS_0 k \cdot A \cdot S_0 \omega \cos^2(\omega t - kx)$$

$$\text{Average power } \langle \text{Power} \rangle = \frac{BS_0^2 A \omega k}{2}$$

$$\langle \text{Intensity} \rangle = \frac{1}{2} BS_0^2 \omega k$$

$$= \frac{1}{2} \Delta \frac{P_0^2}{\rho V}$$

$$V_w = \frac{\omega}{k} = \sqrt{B/\rho}$$

I know these are difficult to learn but try to learn the following formula average intensity.

$$\star \langle I \rangle = \frac{1}{2} BS_0^2 \omega k$$

$$= \frac{1}{2} \Delta \frac{P_0^2}{\rho V \omega}$$

$$= \frac{1}{2} \rho S_0^2 \omega^2 V_w$$

$\star$ If two wave has same velocity $I \propto S_0^2 \omega^2$



Q. If sound of intensity $8340 \times 10^6 \text{ watt/m}^2$ is passing through air.

Find S_0 if frequency = 1000 Hz

$$\rho_{\text{air}} = 1.2 \text{ kg/m}^3, V_{\text{sound}} = 330 \text{ m/s}$$

Sol. $I = \checkmark$

$$\omega = \checkmark$$

$$\rho = \checkmark$$

$$V_w = \checkmark$$

$$S_0 = ? = \checkmark$$

Find S_0 from formula

$$\langle I \rangle = \frac{1}{2} \rho S_0^2 \omega^2 V_w$$

Given

Q. Two sound wave $S_1 = 10 \sin(100\pi t - 4\pi x)$ and $S_2 = 20 \sin(200\pi t - 2\pi x)$ Propagate in a medium with same speed find a & ratio of intensity

$$\text{Sol. } V = \frac{100\pi}{4\pi} = \frac{200\pi}{2\pi a} \quad [v = \frac{w}{k}]$$

$$\Rightarrow a = 4$$

$$I = \frac{1}{2} BS_0^2 \omega k$$

$$[V_w = \frac{\omega}{V_w} \text{ and } k = \frac{\omega}{V_w}]$$

put $k = \frac{\omega}{V_w}$ in I , we get

$$I = \frac{1}{2} BS_0^2 \omega \frac{\omega}{V_w}$$

$$I \propto S_0^2 \omega^2$$

$$\left(\frac{I_1}{I_2} \right) = \left(\frac{10}{20} \right)^2 \times \left(\frac{10\pi}{200\pi} \right)^2$$

$$= \checkmark$$

Standing Wave (Longitudinal)

भाई इसकी maths बिल्कुल standing wave in wave on string जैसी ही है So, इसका load बिल्कुल मत लेना bcz यहाँ result almost पहले जैसे ही है।



When two longitudinal wave of same frequency and amplitude travelling in opposite direction interfere it produce standing wave

$$S_1 = S_0 \sin(kx - \omega t)$$

$$S_2 = S_0 \sin(kx + \omega t)$$

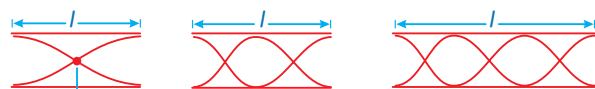
$$S_{\text{net}} = S_0 \sin kx \cos \omega t - S_0 \cos kx \sin \omega t + S_0 \sin kx \cos \omega t + S_0 \cos kx \sin \omega t$$

$$S_{\text{net}} = \boxed{2S_0 \sin kx} \cos \omega t$$

Amplitude

$$\text{Amplitude} = A(x) = \boxed{2S_0 \cos kx}$$

Open Organ Pipe



Displacement Node,
Pressure antinode

$$l = \frac{\lambda}{2} \text{ or } \lambda = 2l$$

$$l = \lambda$$

$$l = \frac{3\lambda}{2} \text{ or } \lambda = \frac{2l}{3}$$

$$f_1 = \frac{v}{\lambda} = \frac{v}{2l}$$

$$f = \frac{v}{l} = 2f_1$$

$$f = \frac{3v}{2l} = 3f_1$$

1st harmonic

2nd harmonic

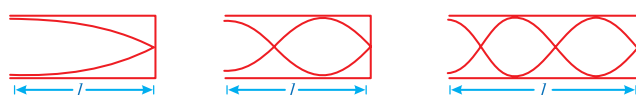
3rd harmonic

Fundamental mode

1st overtone

2nd overtone

Closed Organ Pipe



$$l = \frac{\lambda}{4} \Rightarrow \lambda = 4l$$

$$l = \frac{3\lambda}{4} \Rightarrow \lambda = \frac{4l}{3}$$

$$l = \frac{5\lambda}{4} \Rightarrow \lambda = \frac{4l}{5}$$

$$f = \frac{v}{4l}$$

$$f = \frac{3v}{4l}$$

$$f = \frac{5v}{4l}$$

1st harmonic

3rd harmonic

5th harmonic

Fundamental mode

1st overtone

2nd overtone

Pipe	Fundamental mode	1 st overtone	(n-1) th overtone	Ratio of successive frequency
Open	$f = \frac{v}{2l}$ 1 st Harmonic	$f = \frac{v}{l}$ 2 nd Harmonic	$f = n \frac{v}{2l}$ n^{th} Harmonic	1 : 2 : 3 : 4
Closed	$f = \frac{v}{4l}$ 1 st Harmonic	$f = \frac{3v}{4l}$ 3 rd Harmonic	$f = (2n-1) \frac{v}{4l}$ (2n-1) th Harmonic	1 : 3 : 5 : 7

Where, $v = \sqrt{\frac{\gamma RT}{m}}$

Q. Third overtone of a closed organ pipe is in unison with fourth harmonic of an open organ pipe. Find ratio of their length

Sol. $f_{7(\text{cop})} = f_{4(\text{cop})}$

$$\frac{7v}{4l_1} = \frac{4v}{2l_2} \text{ COP} \rightarrow \text{Closed organ pipe}$$

$$\frac{l_2}{l_1} = \frac{8}{7}$$

OOP - Open organ pipe



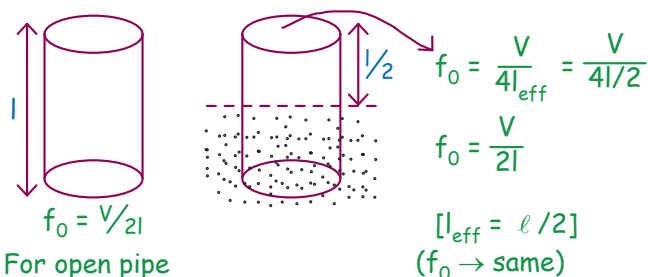
काम का डब्बा

Both open end pipe	One end close
$f_0 = \frac{v}{2l}$	$f_0 = \frac{v}{4l}$
$f_n = \frac{n v}{2l}$ n th harmonic (n-1) overtone	5 th har. = $f_5 = \frac{5v}{4l}$
string case	(2n+1) th harmonic = $f = (2n+1) \frac{v}{4l} = (n^{\text{th}})$ overtone

	Both open end (O.O.P)	One end close (COP)
1 st harmonic	$f_0 = \frac{v}{2l}$ zero overtone	$f_0 = \frac{v}{4l}$ zero overtone
2 nd harmonic	$f_2 = \frac{2v}{2l}$ first overtone	not possible
3 rd harmonic	$f_3 = \frac{3v}{2l}$ 2 nd overtone	$f_3 = \frac{3v}{4l}$ 1 st overtone
3rd overtone formula	4 th harmonic $f_4 = \frac{4v}{2l}$	7 th harmonic $f_7 = \frac{7v}{4l}$

Q. Cylindrical tube open at both end has fundamental freq 'f' in air. Now tube is dipped vertically in water so that half of it is in the water. Fundamental freq of air column is now?

Sol. For closed pipe



Q. For a closed organ pipe if O_2 gas is filled fundamental mode frequency is f_0 at $50^\circ C$

$$f_0 = \frac{V_0}{4l} = \sqrt{\frac{\gamma_1 RT}{m_1}} \frac{1}{4l}, \text{ Answer following parts}$$

Find fundamental mode (FM) freq if gas is replaced by N_2 and ratio of FM.

Sol. $f' = \sqrt{\frac{\gamma_1 RT}{m_1}} \frac{1}{4l}$

$$\frac{(f.m)_{O_2}}{(f.m)_{N_2}} = \frac{(f_0)_{O_2}}{(f_0)_{N_2}} = \sqrt{\frac{m_{N_2}}{m_{O_2}}}$$

Q. 5th harmonic if gas is replaced by N_2

Sol. $N_2 = f'_5 = 5 \frac{V'_0}{4l}$

$$f'_5 = 5 \sqrt{\frac{\gamma_1 RT}{m(N_2)}} \frac{1}{4l}$$

Q. 10th harmonic at $100^\circ C$ if gas is replaced by He

Sol. For O_2 gas F.M = $f_0 = \sqrt{\frac{7}{5} \frac{R \cdot (273 + 50)}{32}}$

$$\text{For He, } f' = \sqrt{\frac{5}{3} \frac{R \cdot (273 + 100)}{4}}$$

Q. Find the fundamental frequency and the first four overtones of a 15 cm pipe

- If the pipe is closed at one end,
- If the pipe is open at both ends,
- How many overtones may be heard by a person of normal hearing in each of the above cases? Velocity of sound in air = 330 ms^{-1}

Sol. For the organ pipe closed at one end, the fundamental frequency of the wave of wavelength λ is given by, $f_0 = \frac{v}{4L}$. The frequency

of i^{th} over tone is given by $f_i = (2i + 1) \times f_0$ where $i = 1, 2, 3 \dots$ etc.

(i) $f_0 = \frac{v}{4L}$ where f_0 = frequency
of the fundamental
 $\Rightarrow f_0 = \frac{330}{4 \times 0.15} = 550 \text{ Hz}$

$$f_1 = 3f_0, f_2 = 5f_0, f_3 = 7f_0, f_4 = 9f_0$$

So, the required frequencies are, 1650, 2750, 3850 and 4950 Hz

(ii) $f_0 = \frac{v}{2L} = \frac{330}{2 \times 0.15} = 1100 \text{ Hz}$

The first four overtones are $2f_0, 3f_0, 4f_0$ and $5f_0$. So, the required frequencies are 2200, 3300, 4400, and 5500 Hz.

(iii) Open pipe $nf_0 = 20,000$

$$n(1100) = 20,000$$

$$n = 18.18 \approx 18$$

Closed pipe

the frequency of the n^{th} overtone is $(2n + 1)f_0$

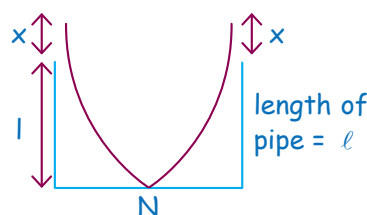
$$\therefore (2n + 1)f_0 = 20000; \text{ or } (2n + 1)550 = 20000$$

$$\text{or } n = 17.68$$

Acceptable value is 17 for both pipes.

End Correction

Effective length in organ pipe is slightly longer than actual length because air vibrates slightly beyond pipes physical end.



For Fundamental Mode For COP

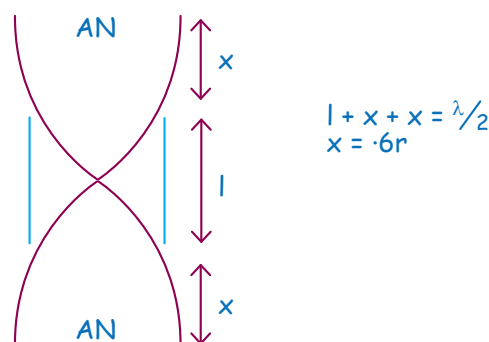
$$l + x = \lambda/4$$

$$\lambda = 4(l + x)$$

$$f_0 = \frac{v}{\lambda} = \frac{v}{4(l + x)}$$

Where $x = .6r$
radius of tube

For Fundamental Mode For OOP

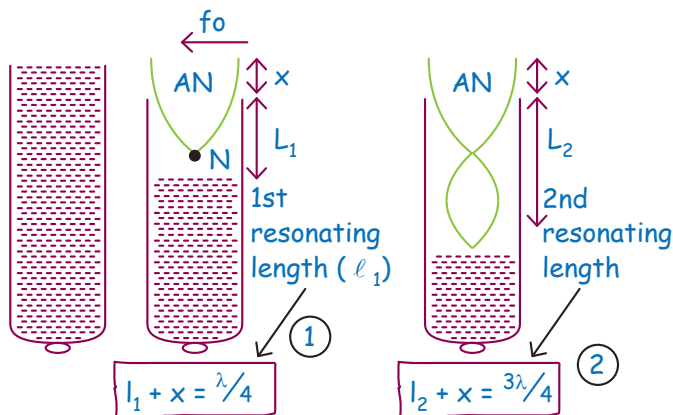


$$l + x + x = \lambda/2$$

$$x = .6r$$

RESONANCE COLUMN TUBE

This experiment is used to find speed of sound in air. It has a tube partially filled with water. A tuning fork is held above open end to create standing waves.



Form 1 & 2

$$\lambda = 2(l_2 - l_1)$$

Velocity of sound is given by

$$V_{\text{sound}} = f_0 \lambda$$

3rd resonating length = l_3

$$l_3 + x = 5\lambda/4$$

Now you can solve according to data

Q. In a resonance column tube exp if 1st resonating length is 40 cm 2nd resonating length is observed at 122 cm find.

(i) End correction/diameter of tube

(ii) Third resonating length

Sol. $l_1 = 40$

$$l_2 = 122$$

$$(i) \quad 40 + x = \lambda/4$$

$$122 + x = 3\lambda/4$$

$$\Rightarrow 82 = \lambda/2$$

$$\lambda = 164$$

$$V_w = f_0 \lambda$$

$$40 + x = \frac{164}{4} \quad [\lambda = 164]$$

$$\text{so } x = 1 \text{ cm}$$

$$x = .6r \Rightarrow 1 \text{ cm} = .6 r$$

$$(ii) \quad l_3 + x = 5\lambda/4$$

$$l_3 + 1 = 5 \times \frac{164}{4}$$

$$l_3 = \checkmark$$

Q. A tube of a certain diameter and of length 48 cm is open at both ends. Its fundamental frequency of resonance is found to be 320 Hz. The velocity of sound in air is 320 m/sec. Estimate the diameter of the tube.

Sol. $V_w = f_0 \lambda$

$$320 = 320 \lambda$$

$$\Rightarrow \lambda = 1 \text{ m} = 100 \text{ cm}$$

For resonance

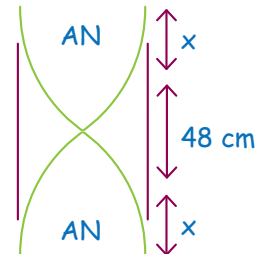
$$x + 48 + x = \lambda/2$$

$$2x + 48 = \frac{100}{2}$$

$$x = 1 \text{ cm} = .6r$$

$$r = \frac{10}{2} \text{ cm}$$

$$\text{diameter} = 2r = \frac{20}{6} = 3.33$$



Beat

When two sound wave of same amplitude and different frequency superimpose then intensity at any point in space varies periodically wrt time. This effect is called beat.

$$f_1 = 500$$

$$f_2 = 503$$

$$\text{Beat} = |f_1 - f_2|$$

$$\text{Beat} = 503 - 500$$

$$\text{frequency} = 3$$

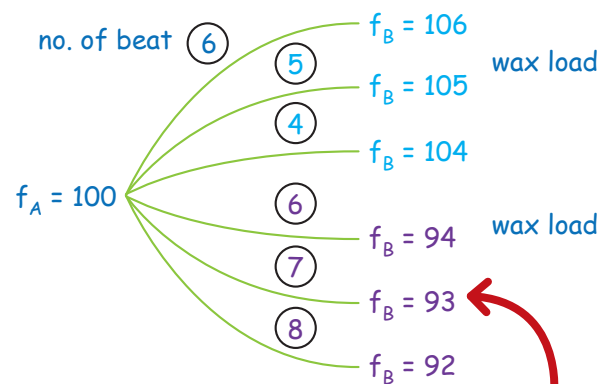
T.F $\Rightarrow$ wax $\Rightarrow$ frequency $\downarrow$

T.F $\Rightarrow$ phile $\Rightarrow$ frequency $\uparrow$ [T.F $\rightarrow$ Turing fork]

Note: On waxing T.F frequency decrease & on philing frequency increase.

Q. Consider a T.F (tuning for k) A of frequency 100 Hz when sounded with T.F B produce 6 beats. If some wax is loaded on B beat frequency increase. Find initial. frequency of T.F of B.

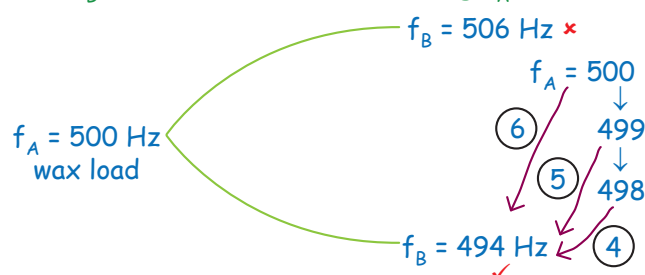
Sol. Frequency of B can be 106 or 94



Ans: $F_B = 94 \text{ Hz}$ (अबे समझ आया ना $F_B = 94$ मे wax load किया तो देखो frequency कम करने पर beat frequency increase हो रही है)

Q. Consider a T.F (tuning fork) A of frequency 500Hz when sounded with T.F B produce 6 beats. If some wax is loaded on A beat frequency decrease to 4. frequency of T.F on B.

Sol. f_B can be 506 or 494. On waxing f_A decrease



Beat reduces with $f_B = 494$ Hz.

Ans: 494

Q. A tuning fork of unknown frequency gives 4 beats with a tuning fork of frequency 310 Hz. It gives the same number of beats on filing. Find the unknown frequency.

Sol. Unknown frequency
= Known frequency $\pm$ Beat frequency
= $310 \pm 4 = 314$ or 306 Hz

Out of these two possible frequencies, one must be the initial value and the other final value. As frequency increases on filing, therefore initial unknown frequency = 306 Hz.

Q. A set of 24 tuning forks is arranged in a series of increasing frequencies. If each fork gives 4 beats per second with the preceding one and the last sounds the octave of the first, find the frequencies of the first and the last forks.

Sol. Let the frequency of the first fork = f
Then the frequency of the second fork = $f + 4$
Frequency of the third fork = $f + 2 \times 4$
Frequency of the fourth fork = $f + 3 \times 4$
Frequency of the 24th fork = $f + 23 \times 4$
But the frequency of the last fork is the octave of the first.
Therefore,
 $2f = f + 23 \times 4$ or $f = 92$ Hz

Thus, frequency of the first fork = $f = 92$ Hz
and frequency of the last fork = $2f = 184$ Hz.

Q. Find velocity of sound in gas where two wave of wavelength 50 cm & 50.4 cm produce 6 $|f_1 - f_2|$

Sol. Beat = $|f_1 - f_2|$
 $v = fx$
 $f_1 - f_2 = 6$
 $\frac{v}{50} - \frac{v}{50.4} = 6$
 $v = \checkmark$

Q. An closed organ pipe(COP) & open organ pipe(OOP) of same length produce 2 beat when they are set into vibration simultaneously in this fundamental mode. If the length of OOP is now half & that of COP is doubled no. of beat produce will be?

Sol. $\left| \frac{v}{4l} - \frac{v}{2l} \right| = 2$ Fundamental mode
frequency $\frac{v}{2l}$

$$\frac{v}{4l} = 2$$

$$\frac{v}{l} = 8$$

When length of OOP is half & that of COP double

$$\frac{v}{4(2l)} - \frac{v}{2(l/2)} = ?$$

$$\frac{v}{8l} - \frac{v}{l} = \frac{7v}{8l}$$

$$\frac{7}{8} \times 8$$

Ans: 7

Q. The first overtone of an open organ pipe beats with the first overtone of a closed organ pipe with a beat frequency of 2.2 Hz. The fundamental frequency of the closed organ pipe is 110 Hz. Find the lengths of the pipes. (Take speed of sound $v_s = 330$ m/s)

Sol. The beats are produced when the waves of different frequencies, resonate with each other.

Let the length of open and closed pipes be l_1 and l_2 respectively.

The frequency of first overtone of open organ pipe is $f_1 = \frac{2v}{2l_1} = \frac{v}{l_1}$

The frequency of first over tone of closed organ pipe is $f_2 = \frac{3v}{4l_2}$

Fundamental frequency of closed organ pipe

$$f = \frac{v}{4l_2} \Rightarrow \frac{v}{4l_2} = 110 \text{ Hz (given)}$$

$$\Rightarrow l_2 = \frac{330}{4 \times 110} = 0.75 \text{ m}$$

As beat frequency = 2.2 Hz, then (if $f_1 > f_2$)

$$\Rightarrow \frac{v}{l_1} - \frac{3v}{4l_2} = 2.2 \Rightarrow \frac{330}{l_1} - \frac{3 \times 330}{4 \times 0.75} = 2.2 \text{ Hz}$$

$$\Rightarrow l_1 = \frac{330}{332.2} = 0.993 \text{ m}$$

$$\text{If } f_2 > f_1, \text{ then } = \frac{3v}{4l_2} - \frac{v}{l_1} = 2.2 \text{ Hz}$$

$$\text{or } \frac{3 \times 330}{4 \times 0.75} - \frac{330}{l_1} = 2.2 \Rightarrow \frac{330}{l_1} = 327.8$$

$$l_1 = 1.007 \text{ m}$$

COHERENT SOURCE

- ★ Two sources are said to be coherent if they produce wave of same frequency with const. phase difference
- ★ Phase difference is const. (independent of time)
- ★ Two independent source cannot be coherent.
- ★ For sustained interference sources must be coherent.
- ★ To create coherent 2 source, it is essential that they should be derived from same common source.

Interference of Sound Wave

इसे अच्छे से कर लेना bcz same as it is mathematic आपको 12th class में wave optics में देखने को मिलेगी।



Interference of Wave

When two waves of equal frequency & equal amplitude traveling in same direction in a medium superimpose this phenomenon is known as interference of wave. When two waves from coherent source superimpose they give definite intensity pattern in definite location at space.

Suppose eqⁿ of two waves are

$$y_1 = A_1 \sin(\omega t - kx)$$

$$y_2 = A_2 \sin(\omega t - kx + \phi)$$

$$\text{Then } y_{\text{net}} = y_1 + y_2 = A_1 \sin(\omega t - kx) + A_2 \sin(\omega t - kx + \phi) \\ = A_{\text{net}} \sin(\omega t - kx + \alpha)$$

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$\text{Intensity} \propto A^2$$

Let $Kx - \omega t = B$

$$\Rightarrow y_{\text{net}} = A_1 \sin B + A_2 \sin(B + \phi)$$

$$= A_1 \sin B + A_2 \sin B \cos \phi + A_2 \cos B \sin \phi$$

$$y_{\text{net}} = (A_1 + A_2 \cos \phi) \sin(\omega t - kx) + (A_2 \sin \phi) \cos(\omega t - kx)$$

$$= a \sin(\omega t - kx) + b \cos(\omega t - kx)$$

$$y_{\text{net}} = a \sin(\omega t - kx) + b \cos(\omega t - kx)$$

$$= \sqrt{a^2 + b^2} \sin(\omega t - kx + \alpha)$$

$$= A_{\text{net}} \sin(\omega t - kx + \alpha)$$

$$a = A_1 + A_2 \cos \phi$$

$$b = A_2 \sin \phi$$

$$A_{\text{net}} = \sqrt{a^2 + b^2} = \sqrt{(A_1 + A_2 \cos \phi)^2 + (A_2 \sin \phi)^2}$$

$$= \sqrt{A_1^2 + A_2^2 \cos^2 \phi + 2A_1A_2 \cos \phi + A_2^2 \sin^2 \phi}$$

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

phase difference

$$\tan \alpha = \frac{b}{a} = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}$$

देख भाई maths गई भाड़ में तू बस ये result पर focus कर

Results

$$y_1 = A_1 \sin(\omega t - kx)$$

$$y_2 = A_2 \sin(\omega t - kx + \phi)$$

$$y_{\text{net}} = A_{\text{net}} \sin(\omega t - kx + \alpha)$$

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$\Delta \phi = \phi = (\theta_2 - \theta_1) \text{ phase diff}$$

$$I_{\text{net}} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

phase difference

$$\text{Intensity} \propto A^2$$

$$\frac{\Delta \phi}{2\pi} = \frac{\Delta x}{\lambda} \quad \text{path diff}$$

$$\Delta \phi \rightarrow \text{phase diff.}$$

$$\Delta \phi = \frac{2\pi}{\lambda} \Delta x$$

Constructive Interference

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$



$$I_{\text{net}} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

कहीं तो कोई condition होगी जहाँ $A_{\text{net}}, I_{\text{net}}$ max. मिले

जहाँ $A_{\text{net}} \rightarrow \text{max.}$ वहीं $I_{\text{net}} \rightarrow \text{max.}$

$$\cos \phi = +1 \quad \boxed{\phi = 2n\pi}$$

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \times 1}$$

$$A_{\text{net}} = A_1 + A_2$$

$$I_{\text{net}} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$\text{If } A_1 = A_2 = A_0 \text{ then } A_{\text{net}} = 2A_0$$

$$\text{If } I_1 = I_2 = I_0 \text{ then } I_{\text{net}} = 4I_0$$

$$\text{For } \cos(\Delta\phi) = 1$$

$$\Delta\phi = 2n\pi, \text{ and we know that } \frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

$$\text{So, } \frac{2n\pi}{2\pi} = \frac{\Delta x}{\lambda} \rightarrow \boxed{\Delta x = n\lambda}$$

Destructive Interference

Similarly

$$A_{\text{net}} \rightarrow \text{min.}$$

$$I_{\text{net}} \rightarrow \text{min.}$$

$$\cos \phi = -1$$

$$\boxed{\phi = (\text{odd})\pi}$$

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 - 2A_1A_2}$$

$$= |A_1 - A_2|$$

$$I_{\text{net}} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\text{If } A_1 = A_2 = A_0 \text{ then } A_{\text{net}} = 0$$

$$\text{If } I_1 = I_2 = I_0 \text{ then } I_{\text{net}} = 0$$

$$\Delta\phi = (\text{odd})\pi = (2n+1)\pi$$

$$\Delta\phi/2\pi = \Delta x/\lambda$$

$$\therefore \Delta x = (\text{odd})\lambda/2$$

Wave



काम का डब्बा

Exams से पहले यह जरूर देख कर जाना।

Final Results

$$A_{\text{net}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$I_{\text{net}} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$\text{Intensity} \propto A^2$$

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda} \quad (\Delta x = \text{path diff. and } \Delta\phi \rightarrow \text{phase diff.})$$

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x \quad I = 4I_0 \cos^2(\Delta\phi/2)$$

For Maxima (CI)

$$A_{\text{net}} = A_1 + A_2$$

$$I_{\text{net}} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$\phi = 2n\pi$$

$$\Delta x = n\lambda$$

$$\text{If } A_1 = A_2 = A_0 \text{ then}$$

$$A_{\text{net}} = 2A_0$$

$$\text{If } I_1 = I_2 = I_0 \text{ then } I_{\text{net}} = 4I_0$$

For Minima (DI)

$$A_{\text{net}} = A_1 - A_2$$

$$I_{\text{net}} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\phi = (\text{odd})\pi$$

$$\Delta x = (\text{odd})\lambda/2$$

$$\text{If } A_1 = A_2 = A_0 \text{ then}$$

$$A_{\text{net}} = 0$$

$$\text{If } I_1 = I_2 = I_0 \text{ then } I_{\text{net}} = 0$$

$$\frac{I_{\text{max}}}{I_{\text{min}}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2}$$

If S_1 & S_2 are not coherent

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$I_{\text{net}} = I_1 + I_2 \quad (\text{because } \cos \phi = 0)$$

Q. If $I_1 = 4I_0$ and $I_2 = I_0$ find $\frac{I_{\text{max}}}{I_{\text{min}}}$

$$\text{Ans. } \frac{I_{\text{max}}}{I_{\text{min}}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = 9 \text{ solve and get}$$

Q. $\frac{I_1}{I_2} = \frac{1}{9}$ find 1. $\frac{A_1}{A_2}$ 2. $\frac{I_{\max}}{I_{\min}}$

Sol. $I \propto A^2$

$$\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$$

$$\frac{A_1}{A_2} = \frac{1}{3}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \left(\frac{\sqrt{\frac{I_1}{I_2}} + 1}{\sqrt{\frac{I_1}{I_2}} - 1} \right)^2$$

$$= \left(\frac{\frac{1}{3} + 1}{\frac{1}{3} - 1} \right)^2 = \left(\frac{16}{4} \right) = 4$$

Q. $\frac{I_{\max}}{I_{\min}} = \frac{16}{9}$ find 1. $\frac{I_1}{I_2}$ 2. $\frac{A_1}{A_2}$

Sol. Sir's solution

$$\left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2 = \frac{16}{9}$$

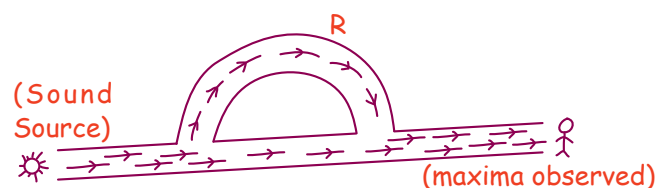
$$\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} = \frac{4}{3}$$

$$\frac{\sqrt{I_1}}{\sqrt{I_2}} = \frac{4+3}{4-3}$$

$$\frac{I_1}{I_2} = 49 = \frac{A_1^2}{A_2^2}$$

$$\frac{A_1}{A_2} = 7$$

Q. Write the equation for maxima detection for observer.



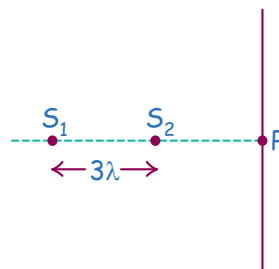
Sol. $\Delta x = n\lambda$

$$\pi R - 2R = n\lambda$$

$$\pi R = n\lambda + 2R; \quad R = \frac{n\lambda}{\pi - 2}$$

Q. Two coherent sources S_1 and S_2 are on x-axis having separation between them equal to 3λ . Find the intensity at origin P. ($I_1 = I_2 = I_0$)

Sol.



$$\Delta x = S_1P - S_2P$$

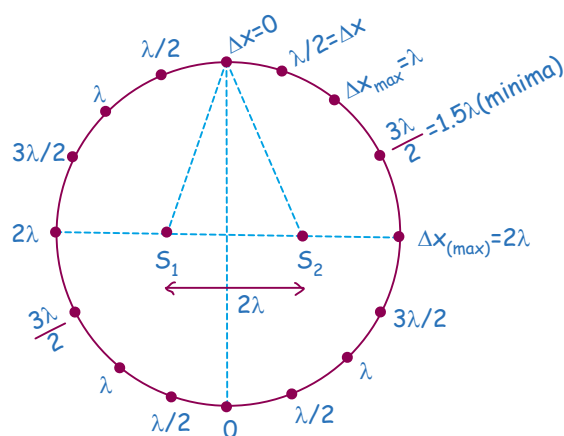
$$\Delta x = 3\lambda \text{ (maxima)}$$

$$(I_P) = 4I_0$$

$$I_1 = I_2 = I_0$$

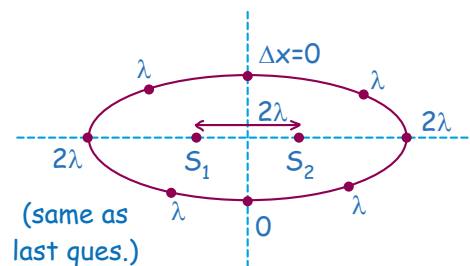
Q. If two coherent sources S_1 and S_2 are on x-axis having separation between them equal to 2λ as shown in figure. Find the number of maxima and minima on the circle.

Sol.

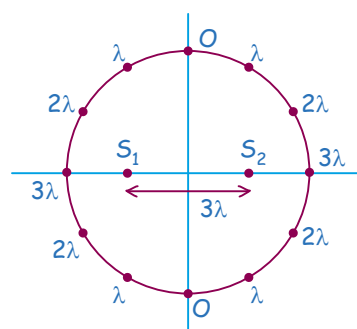


No. of maxima = 8

No. of minima = 8

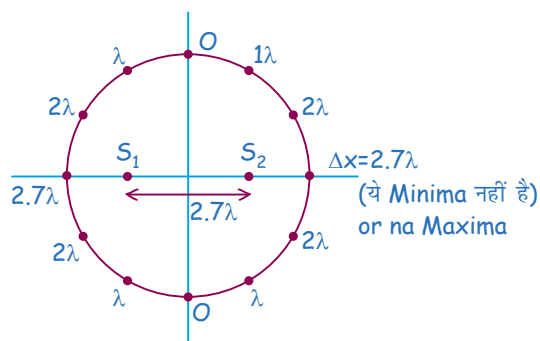


☆



No. of max. on ring = 4 + 8 = 12

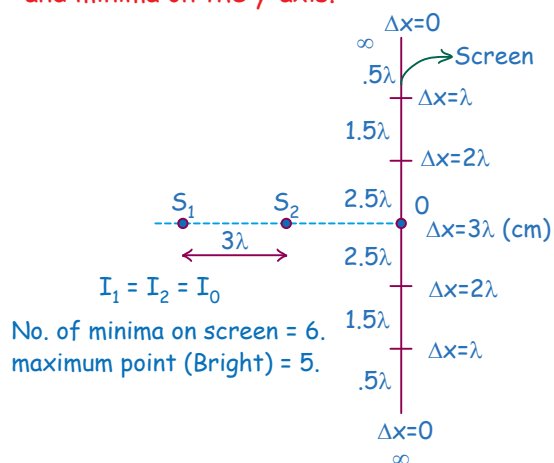
No. of min. = 12



No. of max. = 10

No. of min. = 12

☆ If two coherent sources S_1 and S_2 are on x-axis having separation between them equal to $d = 3\lambda$ as shown in figure. Find the number of maxima and minima on the y-axis.



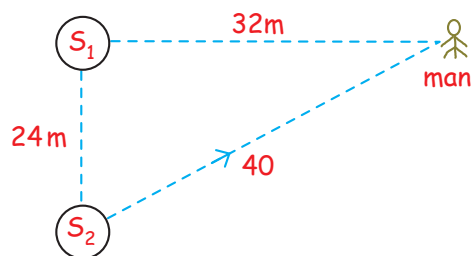
Q. Repeat the above problem for different value of d given in table below.

Sol. $I_1 = I_2 = I_0$

	No. of max. on screen	No. of min. on screen
$d = 3\lambda$	5	6
$d = 3.2\lambda$	6	6
$d = 3.5\lambda$	6	7
$d = 3.73\lambda$	6	8

Q. Two identical coherent sources are fixed as shown in figure. Answer following parts

1. Find value of λ for which man observe minima



Sol. For minima path difference $\Delta x = (\text{odd}) \lambda/2$

$$40 - 32 = (\text{odd}) \lambda/2 = \lambda = 16/\text{odd}$$

$$\lambda = 16, \frac{16}{3}$$

2. Man observe maxima

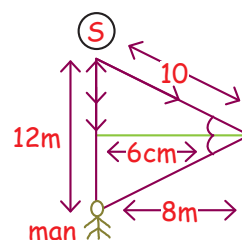
Sol. $\Delta x = n\lambda$

$$8 = n\lambda$$

$$\lambda = 8/n$$

$$\lambda = 8, \frac{8}{2}, \frac{8}{3}$$

Q. Find λ for which man observe minima in figure as shown below.



Sol. $\Delta x = 20 - 12 = (\text{odd}) \lambda/2$

= Path difference

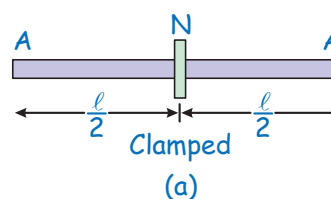
$$\lambda = 16, \frac{16}{3}, \frac{16}{5} \dots$$

LONGITUDINAL VIBRATIONS OF ROD

Rod Free at Both Ends

A rod clamped at the middle and free at both ends, vibrates longitudinally with the node in middle and anti-nodes at the free end. The first two possible modes are shown.

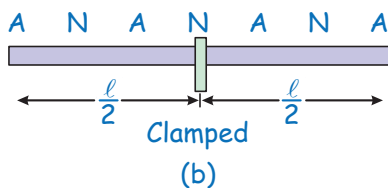
(i) The fundamental mode (lowest frequency note) is shown in figure.(a)



$$\frac{2\lambda_1}{4} = l \Rightarrow \lambda_1 = 2l$$

$$\text{Its frequency is, } f_1 = \frac{v}{\lambda_1} = \frac{v}{2l}$$

(ii) The next mode is shown below.

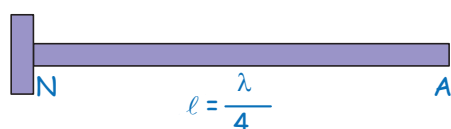


$$\frac{6\lambda_2}{4} = l \Rightarrow \lambda_2 = \frac{2l}{3}$$

$$\text{Its frequency is } f_2 = \frac{v}{\lambda_2} = \left(\frac{v}{2l/3} \right) = 3f_1$$

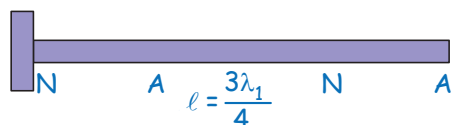
Rod Clamped at One End

Fundamental mode:



$$\text{Frequency: } f_1 = \frac{v}{\lambda_1} = \frac{v}{4l}$$

First overtone:



$$\text{Frequency: } f_2 = \frac{v}{\lambda_2} = 3 \frac{v}{4l} = 3f_1$$

Thus again the overtone frequencies are 3, 5, 7, times that of the fundamental.

DOPPLER'S EFFECT (NOT IN MAINS)

When either the source, the observer, or both are in motion relative to the medium through which the wave propagates, the observed frequency may differ from the frequency emitted by the source. This perceived change in frequency, resulting from the motion of the source or the observer, is referred to as the Doppler effect.

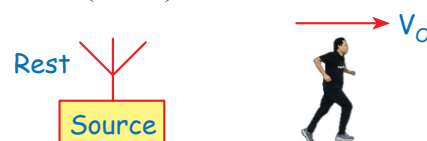
Suppose observer and source are moving with speed V_o and V_s along their line joining and speed of sound is V then apparent frequency observed by observer is calculated by relation.

$$f_{\text{app}} = f_0 \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

Various Cases

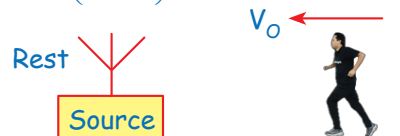
1. Source at rest, observer moves away from source:

$$f_{\text{app}} = f_0 \left(\frac{v - v_o}{v} \right)$$



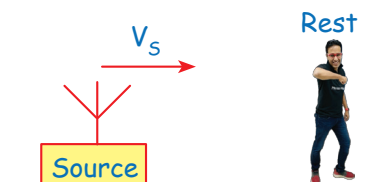
2. Source at rest, observer moves towards from source:

$$f_{\text{app}} = f_0 \left(\frac{v + v_o}{v} \right)$$



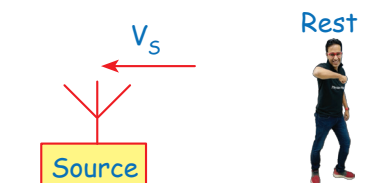
3. Observer at rest, source moves towards observer:

$$f_{\text{app}} = f_0 \left(\frac{v}{v - v_s} \right)$$



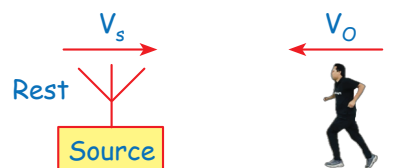
4. Observer at rest, source moves away observer:

$$f_{\text{app}} = f_0 \left(\frac{v}{v + v_s} \right)$$

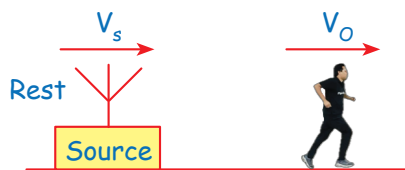


5. Both move and approaching each other:

$$f_{\text{app}} = f_0 \left(\frac{v + v_o}{v - v_s} \right)$$

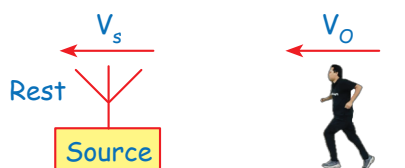


6. Both move and source following the observer:



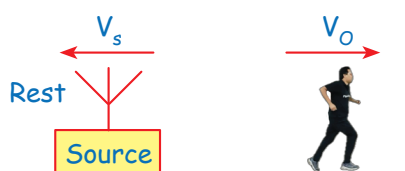
$$f_{app} = f_0 \left(\frac{v - v_o}{v - v_s} \right)$$

7. Both move and observer following the source:



$$f_{app} = \left(\frac{v + v_o}{v + v_s} \right) f_0$$

8. Both moving away from each other:



$$f_{app} = \left(\frac{v - v_o}{v + v_s} \right) f_0$$

Q. A source of sound of frequency 450 cycles/s is moving towards a stationary observer with speed 34 m/s. If the speed of sound is 340 m/s, then the apparent frequency will be

Sol. $n' = n \left(\frac{v}{v - v_s} \right) = 450 \left(\frac{340}{340 - 34} \right) = 500 \text{ cycles / sec}$

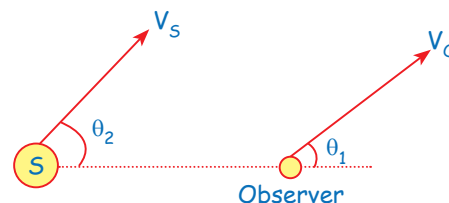
Q. The speed of sound in air at a given temperature is 350 m/s. An engine blows whistle at a frequency of 1200 cps. It is approaching the observer with velocity 50 m/s. The apparent frequency in cps heard by the observer will be

Sol. $n' = n \left(\frac{v}{v - v_s} \right) = 1200 \times \left(\frac{350}{350 - 50} \right) = 1400 \text{ cps}$

If observer and source are moving by making angle θ_1 and θ_2 with their line joining then

$$f_{app} = f_0 \left(\frac{V \pm V_o \cos \theta_1}{V \pm V_s \cos \theta_2} \right)$$

Example

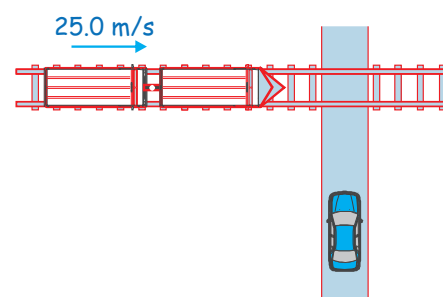


$$f_{app} = f_0 \left(\frac{V - V_o \cos \theta_1}{V - V_s \cos \theta_2} \right)$$

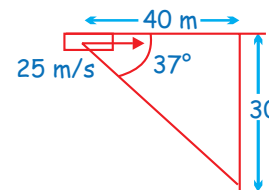
अबे कुछ नही करना बस observer और source की velocity के component करलो और दोनो को join करने वाली line की तरफ वाली velocity formula मे use करना



Q. A train moves at a constant speed of 25.0 m/s towards the intersection shown in figure below. A car is stopped near the intersection, 30.0 m from the tracks. If the train's horn emits a frequency of 160 Hz, what frequency (in Hz) is heard by the passengers in the car, corresponding to the sound emitted by the train when it is at a distance of 40.0 m before the intersection. Take the speed of sound to be 340 m/s.



Sol.



$$\begin{aligned} f &= \frac{c}{c - 25 \cos 37^\circ} \times 160 \\ &= \frac{340}{340 - 25 \times \frac{4}{5}} \times 160 = \frac{340}{320} \times 160 \\ &= 170 \text{ Hz} \end{aligned}$$

QUESTION PRACTICE

Q. Two sources S_1 and S_2 separated by 2.0 m, vibrate according to equation $y_1 = 0.03\sin(\pi t)$ and $y_2 = 0.02\sin(\pi t)$, where y_1 , y_2 and t are in M.K.S. units. They send out waves of velocity 1.5 m/s.

Calculate the amplitude of the resultant motion of the particle collinear with S_1 and S_2 and located at a point.

- (i) To the right of S_2
- (ii) To the left of S_1
- (iii) In the middle of S_1 and S_2

Sol. The phase difference between the two waves is

given by $\phi = \frac{2\pi x}{\lambda}$ where $x = 2.0$ m is the path difference between the two waves at points near to S_1 or S_2 . The resultant amplitude of the superimposed wave is $a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi}$.

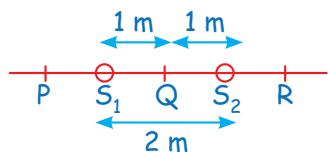
Let P and R be respective points to the left of S_1 and right of S_2 , respectively.

The oscillations y_1 and y_2 have amplitude $a_1 = 0.03$ and $a_2 = 0.02$, respectively. These have equal period $T = 2$ s and same frequency

$$f = \frac{1}{T} = \frac{1}{2} = 0.5 \text{ s}^{-1}.$$

The wavelength of each vibration,

$$\lambda = \frac{v}{f} = \frac{1.5}{0.5} = 3.0 \text{ m}$$



- (i) The path difference for point R to the right of $S_2 = \Delta x = (S_1R - S_2R) = S_1S_2 = 2 \text{ m}$

$$\therefore \text{Phase difference } \phi = \frac{2\pi}{\lambda} x = \frac{2\pi}{3} \times 2.0 = \frac{4\pi}{3}$$

The resultant amplitude for point R is given by

$$\begin{aligned} & \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi} \\ &= \sqrt{\{(0.03)^2 + (0.02)^2 + 2 \times 0.03 \times 0.02 \times \cos(4\pi/3)\}} \end{aligned}$$

Solving, we obtain $a = 0.02565 \text{ m}$.

- (ii) The path difference for point P to the left of S_1 is $\Delta x = S_2P - S_1P = 2.0 \text{ m}$.

Hence, the resultant amplitude for point P to the left of S_1 is 0.02565 m.

- (iii) for a point Q, midway between S_1 and S_2 , the path difference is zero i.e., $\phi = 0$

$$\begin{aligned} a &= \sqrt{a_1^2 + a_2^2 + 2a_1a_2} \\ &= \sqrt{\{(0.03)^2 + (0.02)^2 + 2(0.03)(0.02)\}} \\ &= 0.03 + 0.02 = 0.05 \end{aligned}$$

Q. Find the fundamental frequency and the first four overtones of a 15 cm pipe

- (i) If the pipe is closed at one end,
- (ii) If the pipe is open at both ends,
- (iii) How many overtones may be heard by a person of normal hearing in each of the above cases? Velocity of sound in air = 330 ms^{-1}

Sol. For the organ pipe closed at one end, the fundamental frequency of the wave of wavelength λ is given by, $f_0 = \frac{v}{4L}$. The frequency

of i^{th} over tone is given by $f_i = (2i + 1) \times f_0$ where $i = 1, 2, 3 \dots$ etc.

- (i) $f_0 = \frac{v}{4L}$ where f_0 = frequency of the

$$\text{fundamental} \Rightarrow f_0 = \frac{330}{4 \times 0.15} = 550 \text{ Hz}$$

$$f_1 = 3f_0, f_2 = 5f_0, f_3 = 7f_0, f_4 = 9f_0$$

So, the required frequencies are, 1650, 2750, 3850 and 4950 Hz

- (ii) $f_0 = \frac{v}{2L} = \frac{330}{2 \times 0.15} = 1100 \text{ Hz}$

The first four overtones are $2f_0, 3f_0, 4f_0$ and $5f_0$. So, the required frequencies are 2200, 3300, 4400, and 5500 Hz.

(iii) Open pipe $nf_0 = 20,000$

$$n(1100) = 20,000$$

$$n = 18.18 \approx 18$$

Closed pipe

the frequency of the n^{th} overtone is $(2n + 1)f_0$

$$\therefore (2n + 1)f_0 = 20000; \text{ or } (2n + 1)550 = 20000$$

$$\text{or } n = 17.68$$

Acceptable value is 17 for both pipes.

Q. A tube closed at one end has a vibrating diaphragm at the other end, which may be assumed to be displacement node. It is found that when the frequency of the diaphragm is 2000 Hz, a stationary wave pattern is set up in which the distance between adjacent nodes is 8 cm. When the frequency is gradually reduced, then the stationary wave pattern disappears but another stationary wave pattern reappears at a frequency of 1600 Hz. Calculate

- The speed of sound in air.
- The distance between adjacent nodes at a frequency of 1600 Hz.
- The distance between the diaphragm and the closed end.
- The next lower frequencies at which stationary wave pattern will be obtained.

Sol. The standing wave generated inside the tube closed at one end has the wavelength $\frac{\lambda}{2} = L$ where L is length of the tube. The velocity of the wave in air is given by $v = f\lambda$, where f is the frequency of the sound wave.

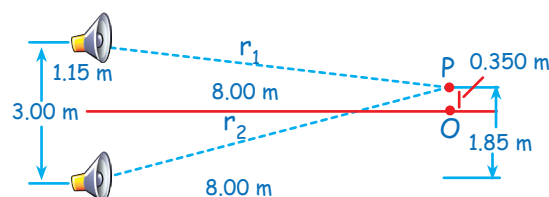
Since the node-to-node distance is $\frac{\lambda}{2}$,

$$\text{or } \Rightarrow \frac{\lambda}{2} = 0.08 \text{ or } \lambda = 0.16 \text{ m}$$

- $v = f\lambda \therefore v = 2000 \times 0.16 = 320 \text{ m/s}$
- $320 = 1600 \times \lambda \text{ or } \lambda = 0.2 \text{ m}$
 $\therefore \text{Distance between nodes} = \frac{0.2}{2} = 0.1 \text{ m} = 10 \text{ cm}$
- Since there are nodes at the ends, the distance between the closed end and the diaphragm must be an integral multiple of $\lambda/2$
 $\therefore L = n\lambda/2 = n \times 0.2/2 = n' \times 0.16/2$
 $\Rightarrow \frac{n}{n'} = \frac{4}{5} \quad (\text{when } n' = 5, n = 4)$
 $L = \frac{n' \times 0.16}{2} = 0.4 \text{ m} = 40 \text{ cm}$
- For the next lower frequency $n = 3$
 $\therefore 0.4 = 3\lambda/2 \text{ or } \lambda = 0.8/3$

$$\text{since } v = f\lambda, f = \frac{320}{0.8/3} = 1200 \text{ Hz}$$

Q. Two identical speakers, placed 3.00 m apart, emit sound waves in sync using the same oscillator, as shown in the diagram below. A person stands at point O, which is 8.00 m from the midpoint of the line joining the two speakers. The person then walks to point P, which is 0.350 m from point O, perpendicular to the line connecting the speakers. At point P, the person hears the first minimum in sound intensity. What is the frequency of the oscillator?



Sol. At point P, the first minimum occurs when the path difference is $\lambda/2$.

So,

- From first speaker to P: $r_1 = \sqrt{(8.00)^2 + (1.15)^2} = 8.08 \text{ m}$

- From second speaker to P:

$$r_2 = \sqrt{(8.00)^2 + (1.85)^2} = 8.21 \text{ m}$$

$$\text{Path difference: } \Delta r = 8.21 - 8.08 = 0.132 \text{ m}$$

$$\text{For first minimum: } \lambda = 2 \times 0.132 = 0.264 \text{ m}$$

Using speed of sound $v = 343 \text{ m/s}$, frequency is:

$$f = \frac{v}{\lambda} = \frac{343}{0.264} \approx 1300 \text{ Hz}$$

Q. A sound wave in air has a frequency of 1665 Hz and a pressure amplitude of $1.13 \times 10^{-3} \text{ Pa}$. Given the density of air as 1.21 kg/m^3 and the speed of sound as 343 m/s , calculate: (a) the displacement and amplitude of the wave, (b) the intensity of the wave.

Sol. (a) Displacement Amplitude (A):

$$\text{Using } A = \frac{\Delta P_{\text{max}}}{2\pi f \rho v} :$$

$$A = \frac{1.13 \times 10^{-3} \text{ Pa}}{2\pi(1665 \text{ Hz})(1.21 \text{ kg/m}^3)(343 \text{ m/s})}$$

$$= 0.26 \times 10^{-9} \text{ m} = 0.26 \text{ nm}$$

(b) Intensity (I): Using $I = \frac{(\Delta P_{\text{max}})^2}{2\rho v}$:

$$I = \frac{(1.13 \times 10^{-3} \text{ Pa})^2}{2(1.21 \text{ kg/m}^3)(343 \text{ m/s})} = 1.5 \text{ nW/m}^2.$$

Q. A sound wave propagates through air and water at 20°C. Given the density of air as 1.20 kg/m³, speed of sound in air as 350 m/s, density of water as 1000 kg/m³, and speed of sound in water as 1512 m/s, calculate:

- The ratio of the pressure amplitude in water to that in air if the intensities and frequencies are equal.
- The ratio of the intensity in water to that in air if the pressure amplitudes are equal.

Sol. (a) Ratio of Pressure Amplitudes $\left(\frac{\Delta P_{\text{water}}}{\Delta P_{\text{air}}} \right)$ for Equal Intensities:

Intensity is given by $I = \frac{(\Delta P_{\text{max}})^2}{2\rho v}$. For equal intensities ($I_{\text{water}} = I_{\text{air}}$) and frequencies:

$$\frac{(\Delta P_{\text{water}})^2}{2\rho_{\text{water}} v_{\text{water}}} = \frac{(\Delta P_{\text{air}})^2}{2\rho_{\text{air}} v_{\text{air}}}$$

$$\frac{\Delta P_{\text{water}}}{\Delta P_{\text{air}}} = \sqrt{\frac{\rho_{\text{water}} v_{\text{water}}}{\rho_{\text{air}} v_{\text{air}}}} = \sqrt{\frac{(1000)(1512)}{(1.20)(350)}} = 60$$

(b) Ratio of Intensities $\left(\frac{I_{\text{water}}}{I_{\text{air}}} \right)$ for equal pressure Amplitudes:

If $\Delta P_{\text{water}} = \Delta P_{\text{air}}$, then:

$$\frac{I_{\text{water}}}{I_{\text{air}}} = \frac{\rho_{\text{air}} v_{\text{air}}}{\rho_{\text{water}} v_{\text{water}}} = \frac{(1.20)(350)}{(1000)(1512)} = \frac{1}{3600}$$